

# Full System Design, Page Architectures & Data Specification Manual

Satellite Remote Sensing × Agronomic AI × In-Situ IoT Telemetry × Parametric Underwriting

|                              |                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------|
| Document Version:            | Release v1.0.0 (Production Master)                                                |
| Target Infrastructure:       | Vercel Edge (Next.js 16) + Render Core API + PostgreSQL/PostGIS                   |
| Primary Demonstration Pilot: | Plot #204 (Rajesh Mondal, Nadia District, West Bengal • Aman Paddy)               |
| Operational Philosophy:      | Detect Early → Intervene Precisely → Preserve Yield → Settle Automatically        |
| Supported Languages:         | 7 Regional Languages (English, Bengali, Hindi, Telugu, Tamil, Kannada, Malayalam) |
| Security & Access:           | Role-Based Access Control (Farmer, Insurer, System Admin, Field Agent)            |
| Generated Timestamp:         | September 2026 • Official Platform Release Document                               |

**Executive Architectural Abstract:** AgriSure Intelligence is a closed-loop cyber-physical agriculture resilience and parametric underwriting platform. By pairing European Space Agency Sentinel-2 optical imagery (10m multispectral) and Sentinel-1 C-band SAR radar cross-polarization with real-time in-situ ESP32 telemetry (volumetric root soil moisture, ambient micro-weather), the system computes localized Shannon Entropy Crop Health Factor (CHF) indices. Rather than merely waiting for catastrophic loss, the platform executes a 6-stage operational resilience cycle: it detects stress days before visual foliar wilting, prescribes bio-inspired GA-PSO nutrient/irrigation remedies, actuates solar pumps, verifies post-intervention vegetative rebound, and automatically settles residual losses via smart parametric contracts with direct-to-bank DBT settlement—entirely eliminating surveyor delays and fraudulent claim friction.

# 1. System Architecture & 4-Pillar Foundation

AgriSure Intelligence is architected across four interlocking technological pillars that function as a unified operational loop:

|                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pillar 1: Satellite Remote Sensing</b> <ul style="list-style-type: none"><li>• Sentinel-2 MSI: 10m spatial resolution across 13 spectral bands.</li><li>• Sentinel-1 SAR: Dual-pol (VH/VV) C-band radar penetrative of monsoon clouds.</li><li>• Computed Vegetation Indices: NDVI, NDRE, LSWI, EVI, and radar roughness ratios.</li><li>• 4-year localized pixel baseline comparison against historical Kharif averages.</li></ul>        | <b>Pillar 2: In-Situ IoT Telemetry</b> <ul style="list-style-type: none"><li>• ESP32 MCU nodes reporting at 3-second telemetry polling cycles.</li><li>• Dual capacitance probes: Root Zone (15 cm) and Deep Sub-surface (30 cm).</li><li>• Ambient DHT22 micro-climate stations &amp; digital N-P-K soil conductivity probes.</li><li>• Remote hardware actuation: Bi-directional relays controlling solar micro-irrigation.</li></ul>            |
| <b>Pillar 3: Agronomic Intelligence (AI)</b> <ul style="list-style-type: none"><li>• 38-Class PlantVillage Computer Vision classifier (92.9% precision).</li><li>• GA-PSO Hybrid bio-inspired optimizer for precise NPK/water prescriptions.</li><li>• Shannon Entropy Crop Health Factor (CHF) combining 5 normalized sub-indices.</li><li>• Early stress warning: Identifies physiological stress 5-7 days before visual wilting.</li></ul> | <b>Pillar 4: Parametric FinTech &amp; Underwriting</b> <ul style="list-style-type: none"><li>• Zero-paperwork index-based trigger specification (e.g. CHF ≤ 0.580 for 14 days).</li><li>• Automated DBT bank payout calculation via PMFBY subsidy framework (80% subsidy).</li><li>• 11-Section Cryptographic Claim Evidence Dossiers for re-insurers.</li><li>• Chained SHA-256 Merkle audit trail ensuring zero dispute or corruption.</li></ul> |

## The 6-Stage Closed-Loop Resilience Lifecycle

| Stage | Stage Name | Core Technology & Data Source                                | Operational Outcome                                              |
|-------|------------|--------------------------------------------------------------|------------------------------------------------------------------|
| 01    | OBSERVE    | Sentinel-2 MSI (10m) + Sentinel-1 SAR + ESP32 IoT probes     | Raw multi-spectral and in-situ moisture time-series ingestion    |
| 02    | DIAGNOSE   | Shannon Entropy CHF index + 38-class leaf computer vision    | Detects physiological stress days prior to visual canopy wilting |
| 03    | PRESCRIBE  | GA-PSO bio-inspired heuristic multi-objective optimizer      | Generates calibrated N-P-K dosages and micro-irrigation volume   |
| 04    | ACTUATE    | Solar pump remote relay switch + voice mobile advisories     | Executes 45-min drip cycle or targeted organic bio-control spray |
| 05    | VERIFY     | Bi-weekly satellite return passes + root capacitance sensors | Validates post-intervention vegetative recovery trajectory       |
| 06    | INSURE     | Parametric smart contract index breach trigger               | Automatic RBI NEFT payout to DBT account if shock exceeds remedy |

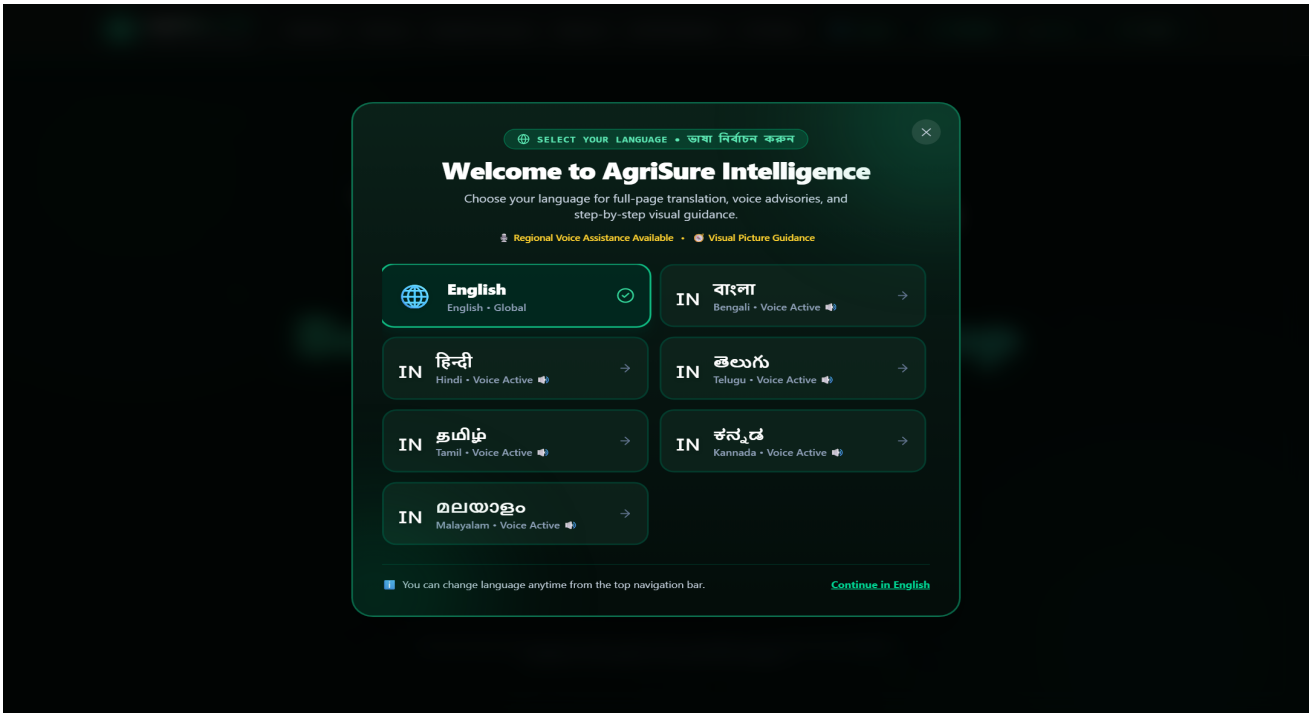
## 2. Detailed Page Designs, Data Requirements & Field Specifications

### 2.1. Public Landing Page & Closed-Loop Cockpit

Route: /

Authorized RBAC Persona: Public Universal (Role Adaptive) | Tier: Presentation & Discovery

Language Support: English (or 7 Regional Languages for Farmers)



#### Functional Purpose & UX Design:

Serves as the high-impact portal showcasing the 4-pillar architectural foundation. Features dynamic 3D elevation cards, live telemetry ticker, benchmark KPI performance targets, the 6-Stage Resilience Cycle, role-filtered CTA buttons, and a live demonstration card of the Nadia pilot farm. Adapts dynamically: if role is Farmer, displays regional language guidance and restricts actions exclusively to farmer portals.

#### Data Required & Technical Pipeline:

- **Real-Time Telemetry Tick:** Polls `/api/iot/devices/iot-dev-1` every 3s to retrieve volumetric soil moisture and pump relay status.
- **Agronomic Research Benchmarks:** Static calibration matrices benchmarked against PlantVillage (38 foliar disease classes, 92.9% accuracy).
- **Pilot Farm Reference State:** Pre-seeded pilot record for Plot #204 Nadia District, Aman Paddy, showing CHF 0.71 vigour status.
- **Role-Based CTA Dynamic State:** Inspects `agrisure_user_role` cookie. Renders Farmer Hub & Apply Insurance for FARMER; Underwriter Desk for INSURER.

#### Field Specification & Data Schema:

| Field / Metric Name | Data Type | UI Control | Req? | Validation Rules & Technical Logic            | Sample Production Value |
|---------------------|-----------|------------|------|-----------------------------------------------|-------------------------|
| Disease Taxonomy    | Integer   | KPI Card   | No   | Calibrated taxonomy classes from PlantVillage | 38 classes              |

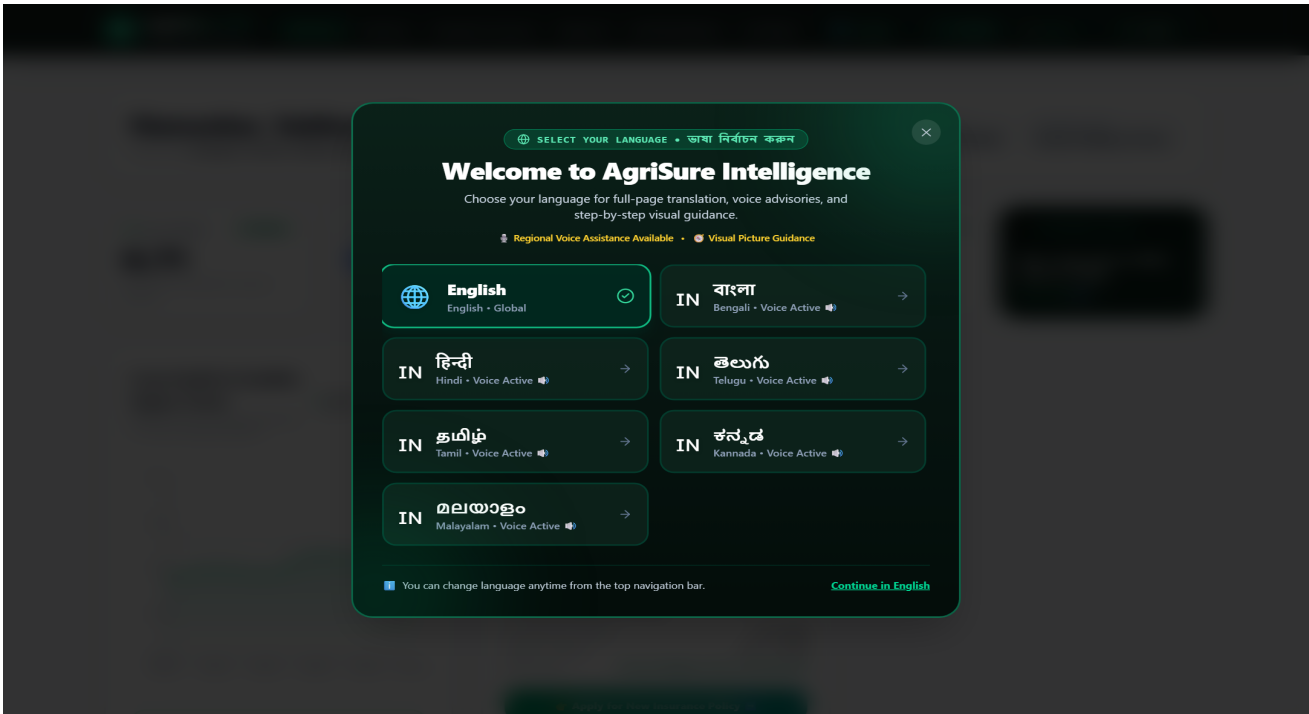
|                      |                    |                  |    |                                                       |                       |
|----------------------|--------------------|------------------|----|-------------------------------------------------------|-----------------------|
| Model Precision      | Float (Percentage) | KPI Card         | No | Accuracy score across foliar blight & brown spot      | 92.9%                 |
| CHF Correlation (r)  | Float              | KPI Card         | No | Pearson r correlation against actual harvest yields   | 0.85                  |
| Claim Target Speed   | String (Range)     | KPI Card         | No | Benchmark resolution turnaround vs 6-9 mo traditional | 30–45 days            |
| IoT Polling Interval | Integer (Seconds)  | KPI Card         | No | ESP32 transmission cycle frequency                    | 3s                    |
| Pilot Plot ID        | String             | Live Unit Badge  | No | Cadastral identifier for primary demonstration farm   | Plot #204 (Baganchra) |
| Pilot CHF Health     | Float (0.00-1.00)  | Status Indicator | No | Composite Crop Health Factor (Normal: 0.76)           | 0.71 (Optimal)        |
| Pilot Soil Moisture  | Float (Percentage) | Live Readout     | No | Volumetric Water Content (VWC) at 15cm                | 58.0% VWC             |
| Weather Risk State   | String (Enum)      | Pill Badge       | No | Values: LOW, MODERATE, HIGH, SEVERE                   | Low (Monsoon Inflow)  |

## 2.2. Farmer Agronomic Resilience Hub

Route: /farmer/dashboard

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Core Operational  
Workspace

Language Support: 7 Regional Languages with  
Audio Synthesis



### Functional Purpose & UX Design:

Central daily management console for smallholder farmers. Built with a 4-step visual pictorial guide and regional voice synthesis for low-literacy users. Displays plain-language vigour scores, volumetric root soil moisture, disease status, prescribed actions, 4-day weather forecasts, and an interactive 1-click pump toggle actuating physical field micro-drip relays.

### Data Required & Technical Pipeline:

- **Farm Master Record:** GET /api/farms/farm-plot-204 retrieves boundary, acreage, crop type, and cadastral registration.
- **In-Situ Sensor Stream:** GET /api/farms/farm-plot-204/sensors retrieves real-time ESP32 soil moisture (58.0%) and soil temp (28.4°C).
- **Micro-Climate Forecast:** GET /api/farms/farm-plot-204/weather provides 4-day precipitation forecasts (4mm, 12mm, 24mm, 0mm).
- **Pump Relay Actuation:** POST /api/iot/devices/iot-dev-1/toggle sends payload { action: 'toggle' } to flip physical pump relay.
- **Speech Synthesis Engine:** Web Audio API generates 2-tone melodic chime; SpeechSynthesis speaks localized advisory in bn-IN, te-IN, etc.

### Field Specification & Data Schema:

| Field / Metric Name | Data Type         | UI Control        | Req? | Validation Rules & Technical Logic         | Sample Production Value        |
|---------------------|-------------------|-------------------|------|--------------------------------------------|--------------------------------|
| Farmer Name         | String            | Greeting Header   | No   | Authenticated farmer identity from profile | Subhash Biswas / Rajesh Mondal |
| Selected Plot       | String            | Dropdown Selector | Yes  | Plot ID matching registered landholdings   | Plot #204 (Baganchra)          |
| Crop Health (CHF)   | Float (0.00-1.00) | Health Card (■)   | No   | Shannon entropy composite vigour index     | 0.71 (Optimal Vigour)          |

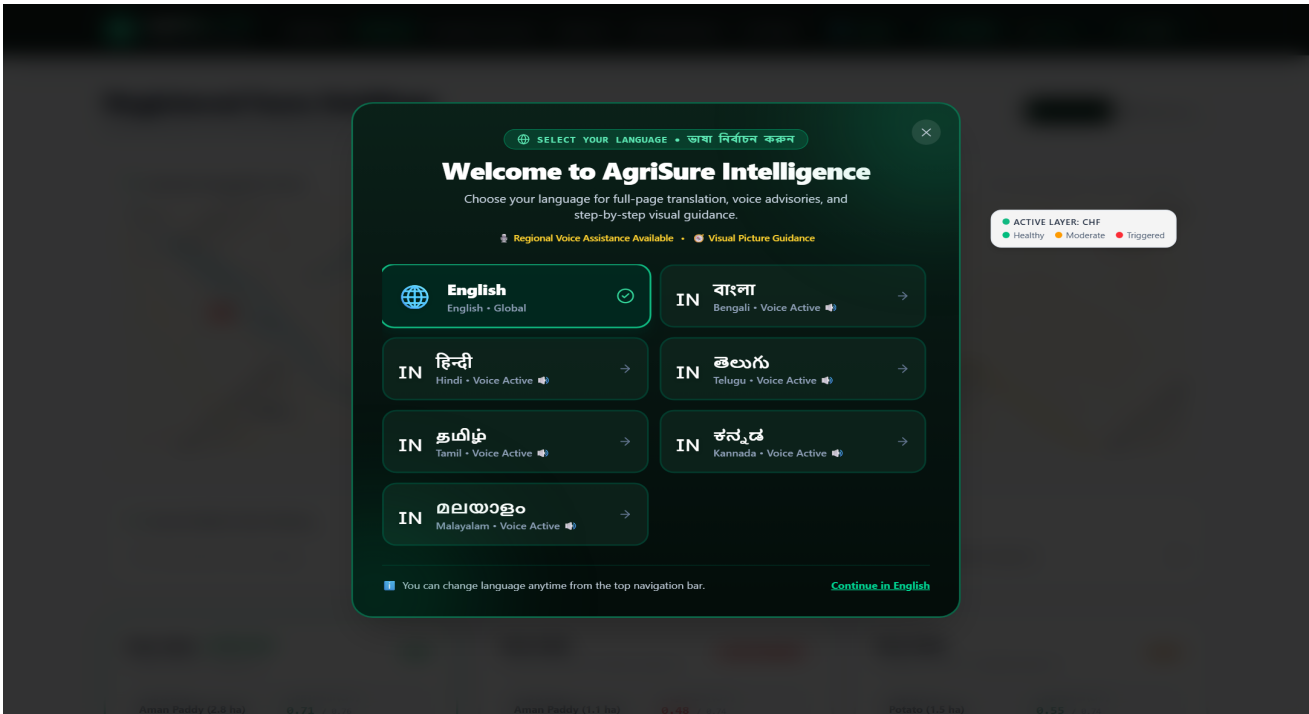
|                            |                  |                    |     |                                                   |                              |
|----------------------------|------------------|--------------------|-----|---------------------------------------------------|------------------------------|
| <b>Soil Moisture (VWC)</b> | Float (%)        | Moisture Card (■)  | No  | Volumetric water content from ESP32 capacitance   | 58.0%                        |
| <b>Weather Risk</b>        | String (Enum)    | Weather Card (■)   | No  | Computed synoptic storm/drought risk level        | Low Risk (Monsoon Steady)    |
| <b>Disease Threat</b>      | String           | Disease Card (■)   | No  | Foliar blight inference from satellite/CV scan    | Canopy Clear (0% Blight)     |
| <b>Prescribed Action</b>   | String           | Action Card (■)    | No  | Daily agronomic prescription in regional language | Micro-drip pulse on Friday   |
| <b>Pump Toggle Switch</b>  | Boolean          | Button Control     | Yes | Toggles state: TRUE (Running) or FALSE (Idle)     | PUMP RUNNING / STANDBY       |
| <b>Health Trend Series</b> | Array of Objects | Recharts LineChart | No  | 6-week historical CHF progression vs normal 0.76  | [Week 1: 68 ... Current: 71] |

### 2.3. My Farms Portfolio & Satellite Index Map

Route: /farmer/farms

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Geospatial Farm Management

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Geospatial landholding explorer allowing farmers to inspect all registered plots. Renders interactive satellite overlay boundaries (Leaflet GIS), area acreages, crop types, planting dates, and multi-spectral vegetation indices (NDVI, NDRE, LSWI, VH/VV backscatter).

Data Required & Technical Pipeline:

- **Plots GeoJSON Collection:** GET /api/farms returns GeoJSON polygon coordinates, soil characteristics, and assigned Insurance Unit code.
- **Multispectral Tile Endpoint:** GET /api/farms/[id]/satellite provides Sentinel-2 B4, B8, B11 reflectance ratios and cloud mask metadata.
- **Entropy CHF Calculation:** GET /api/farms/[id]/chf breaks down the 5 weighted sub-indices driving current health score.

Field Specification & Data Schema:

| Field / Metric Name  | Data Type           | UI Control      | Req? | Validation Rules & Technical Logic                | Sample Production Value   |
|----------------------|---------------------|-----------------|------|---------------------------------------------------|---------------------------|
| Plot Identifier      | String (Unique)     | Plot Selector   | Yes  | Unique cadastral plot key (e.g. farm-plot-204)    | Plot #204 (Baganchra)     |
| Land Acreage (ha)    | Float               | Numeric Badge   | Yes  | Total cultivable plot area in hectares            | 2.8 ha                    |
| Crop Cultivar        | String              | Text Input      | Yes  | Active crop name (e.g. Aman Paddy, Jute, Potato)  | Aman Paddy (Swarna Sub-1) |
| Planting Sowing Date | Date (YYYY-MM-DD)   | Date Picker     | Yes  | Kharif germination / transplanting date           | 2026-06-15                |
| Normalized NDVI      | Float (-1.0 to 1.0) | Telemetry Gauge | No   | Near-infrared vs Red reflectance: (B8-B4)/(B8+B4) | 0.68 (Healthy Canopy)     |

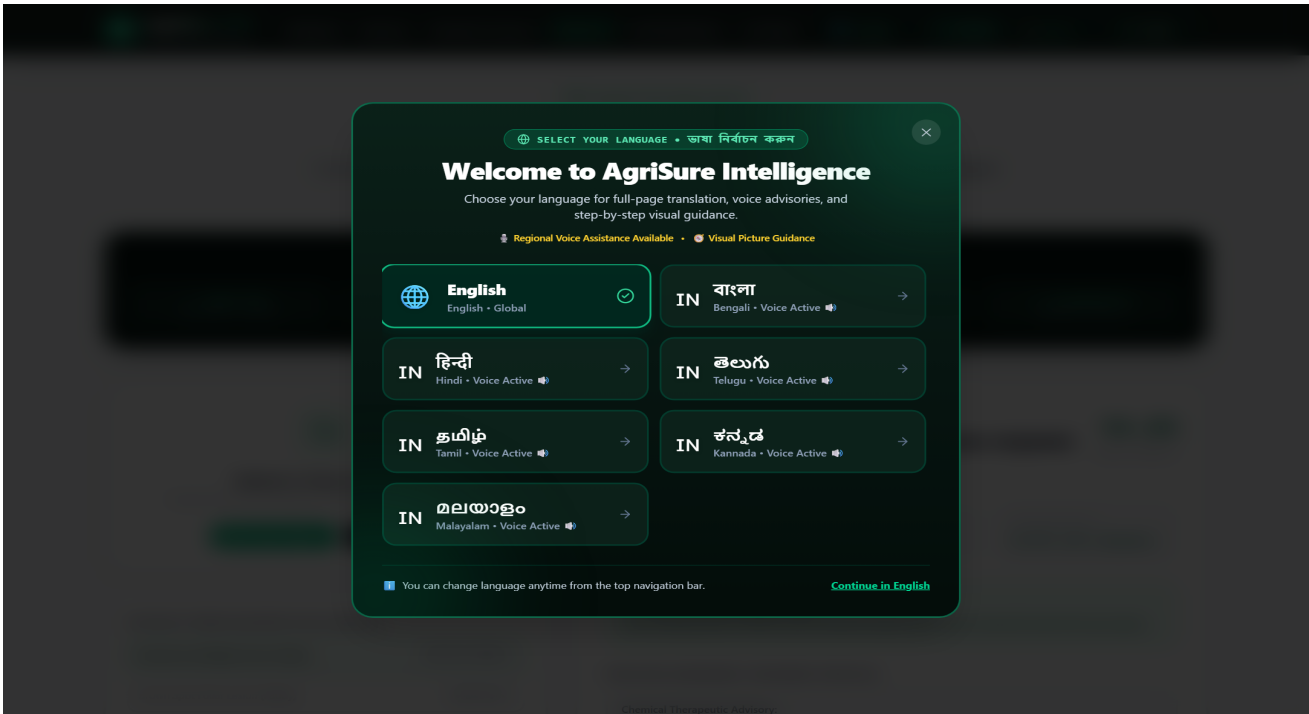
|                         |                     |                 |    |                                                     |                            |
|-------------------------|---------------------|-----------------|----|-----------------------------------------------------|----------------------------|
| Soil Water Index (LSWI) | Float (-1.0 to 1.0) | Telemetry Gauge | No | Canopy moisture index: (B8-B11)/(B8+B11)            | 0.42 (Adequate Moisture)   |
| SAR VH Backscatter      | Float (dB)          | Radar Metric    | No | Sentinel-1 cross-polarization volumetric scattering | -14.2 dB                   |
| Insurance Unit (IU)     | String (Code)       | Readonly Code   | No | Government PMFBY administrative cluster code        | WB-NAD-001 (Nadia Central) |

## 2.4. Computer Vision Leaf Disease AI

Route: /farmer/disease

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Edge Agronomic  
Diagnostics

Language Support: 7 Regional Languages



### Functional Purpose & UX Design:

Mobile-optimized diagnostic portal where farmers take a photo of an affected leaf or upload an image. The deep convolutional neural network classifies the foliar condition across 38 crop disease categories, provides a model confidence rating, severity level, and immediate scientific/organic chemical treatment prescriptions with exact sprayer dosages.

### Data Required & Technical Pipeline:

- **Diagnostic Inference API:** POST /api/disease/predict receives multipart foliar image file or base64 data string.
- **Disease Classification Model:** ResNet-50 / MobileNet backbone calibrated on 54,305 PlantVillage foliar samples across 14 crop species.
- **Treatment Repository:** Database lookup linking detected pathogen to registered CIBRC (Central Insecticides Board) approved treatments.

### Field Specification & Data Schema:

| Field / Metric Name     | Data Type         | UI Control             | Req? | Validation Rules & Technical Logic                 | Sample Production Value            |
|-------------------------|-------------------|------------------------|------|----------------------------------------------------|------------------------------------|
| Leaf Photo Upload       | Binary (JPEG/PNG) | File Dropzone / Camera | Yes  | Max file size: 10 MB. Foliar crop image required   | IMG_rice_leaf_0842.jpg             |
| Associated Farm Plot    | String            | Dropdown Selector      | Yes  | Plot context where the leaf was sampled            | Plot #204 (Baganchra)              |
| Detected Diagnosis      | String            | Result Card            | No   | Predicted disease name or Healthy confirmation     | Rice Brown Spot (Bipolaris oryzae) |
| Confidence Rating       | Float (%)         | Confidence Bar         | No   | Softmax probability distribution score ≥ 85%       | 94.6% Confidence                   |
| Severity Classification | String (Enum)     | Pill Badge             | No   | Values: INCIPIENT (L1), MODERATE (L2), SEVERE (L3) | Stage 1 (Incipient Lesions)        |

|                         |        |                  |    |                                                   |                                      |
|-------------------------|--------|------------------|----|---------------------------------------------------|--------------------------------------|
| Chemical Remedy         | String | Prescription Box | No | Fungicide / bactericide dosage per liter of water | Hexaconazole 5% SC @ 2.0 ml/L        |
| Bio-Control Alternative | String | Prescription Box | No | Organic microbial spray recommendations           | Trichoderma viride @ 5g/L foliar     |
| Application Timing      | String | Schedule Alert   | No | Optimal diurnal window for foliar spray           | Apply late afternoon (after 4:30 PM) |

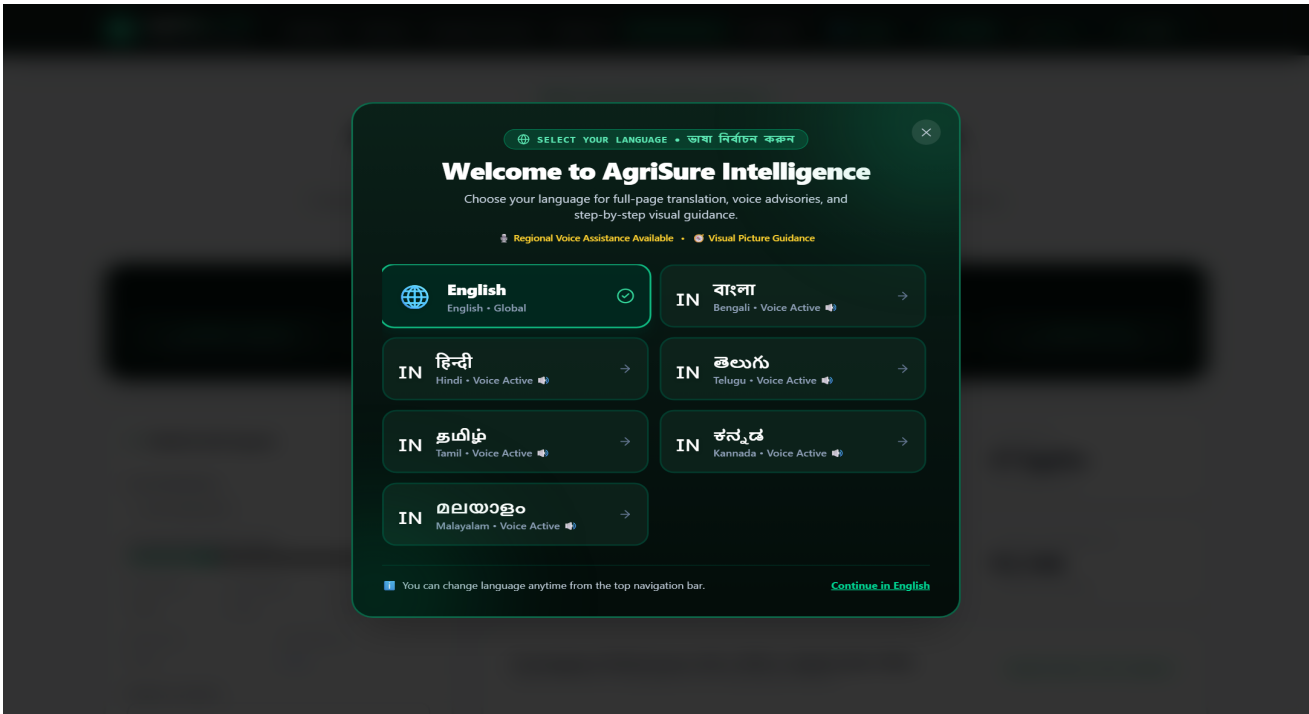
2.5. GA-PSO Bio-Inspired Nutrient Optimizer

Route: /farmer/optimization

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Bio-Inspired

Language Support: 7 Regional Languages

Agronomic Optimization



Functional Purpose & UX Design:

Hybrid heuristic optimizer combining Genetic Algorithms (global exploratory search) and Particle Swarm Optimization (local exploitation) to balance fertilizer investment against maximum attainable yield. Minimizes nitrate leaching and nitrogen volatilization while respecting soil chemical equilibria.

Data Required & Technical Pipeline:

- **Optimization Solver API:** POST /api/optimization/run sends soil macronutrient baseline, crop growth stage, and acreage.
- **Soil Chemical Baseline:** Retrieves current soil nitrogen (180 kg/ha), phosphorus (24 kg/ha), and potassium (145 kg/ha).
- **Objective Function Evaluation:** Calculates pareto front minimizing input expenditure while maximizing crop vigour curve.

Field Specification & Data Schema:

| Field / Metric Name  | Data Type       | UI Control   | Req? | Validation Rules & Technical Logic                    | Sample Production Value |
|----------------------|-----------------|--------------|------|-------------------------------------------------------|-------------------------|
| Crop Phenology Stage | String (Enum)   | Dropdown     | Yes  | Values: VEGETATIVE, PANICLE_INITIATION, GRAIN_FILLING | Panicle Initiation      |
| Available Soil N     | Float (kg/ha)   | Number Input | Yes  | Soil test / sensor nitrogen: range 50 - 400 kg/ha     | 180 kg/ha               |
| Available Soil P     | Float (kg/ha)   | Number Input | Yes  | Available Olsen phosphorus: range 5 - 80 kg/ha        | 24 kg/ha                |
| Available Soil K     | Float (kg/ha)   | Number Input | Yes  | Available ammonium acetate potassium: range 50 - 500  | 145 kg/ha               |
| Optimization Weight  | Float (0.0-1.0) | Range Slider | Yes  | Balance: 0.0 (Cost Min) to 1.0 (Yield Max)            | 0.70 (Yield Priority)   |
| Recommended Urea     | Float (kg/ha)   | Output Card  | No   | Calibrated 46-0-0 nitrogen fertilizer requirement     | 65.0 kg/ha              |

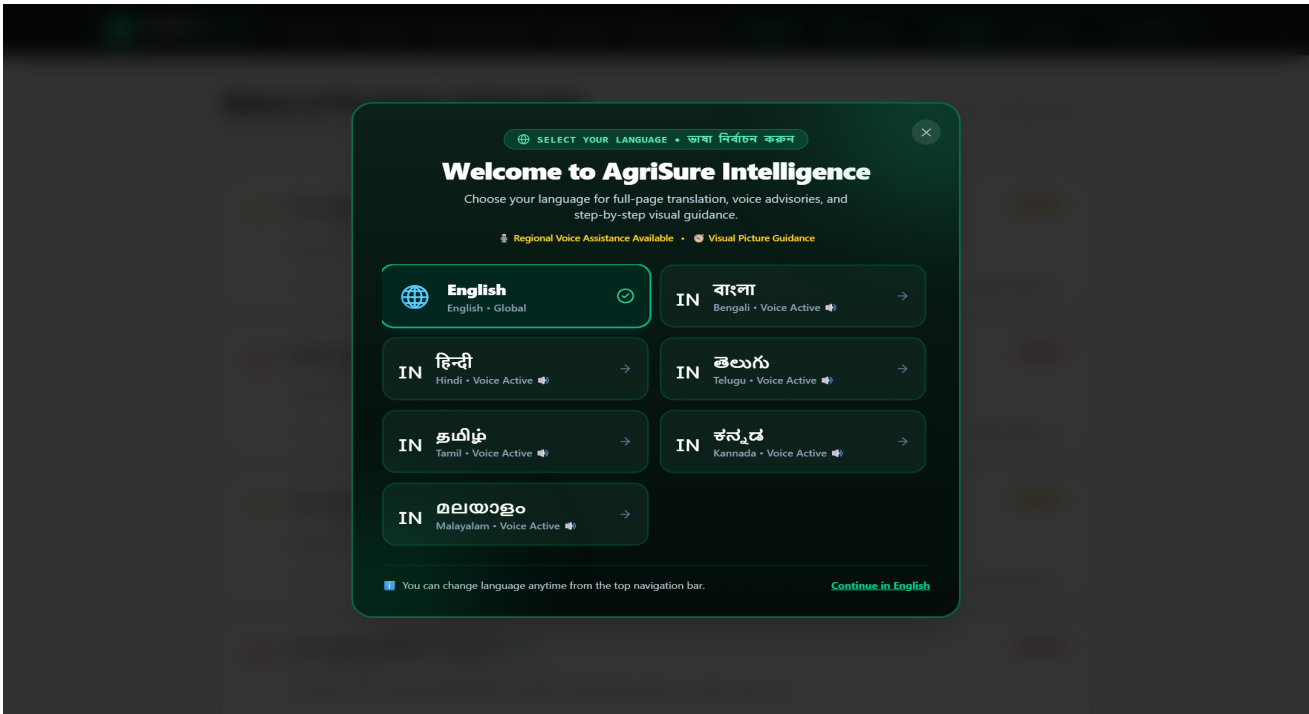
|                    |               |                |    |                                             |                        |
|--------------------|---------------|----------------|----|---------------------------------------------|------------------------|
| Recommended DAP    | Float (kg/ha) | Output Card    | No | Di-ammonium Phosphate 18-46-0 dosage        | 35.0 kg/ha             |
| Recommended MOP    | Float (kg/ha) | Output Card    | No | Muriate of Potash 0-0-60 dosage             | 25.0 kg/ha             |
| Target Water Depth | Float (mm)    | Output Card    | No | Calculated irrigation pulse volume          | 45 mm                  |
| Yield Increment    | Float (%)     | Metric Callout | No | Projected percentage gain over control plot | +14.2% Estimated Yield |

## 2.6. In-Situ IoT Sensor Streams & Telemetry Alerts

Route: /farmer/alerts

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Real-Time Telemetry & Alerting

Language Support: 7 Regional Languages



### Functional Purpose & UX Design:

Live streaming telemetry monitor showing real-time readings from in-field sensor probes and solar weather stations. Generates prioritized alert cards (Critical, Warning, Info) with timestamps and allows farmers to acknowledge alerts or trigger automated pump mitigation with one click.

### Data Required & Technical Pipeline:

- Alerts Stream Endpoint:** GET /api/farms/{id}/alerts lists unacknowledged and historical physiological stress warnings.
- Live Ingestion Telemetry:** GET /api/iot/devices streams active probe battery voltage, signal RSSI, and sensor values.
- Alert Acknowledgment:** POST /api/farms/{id}/alerts/{alertId}/ack updates status from UNREAD to ACKNOWLEDGED.

### Field Specification & Data Schema:

| Field / Metric Name     | Data Type      | UI Control        | Req? | Validation Rules & Technical Logic                   | Sample Production Value           |
|-------------------------|----------------|-------------------|------|------------------------------------------------------|-----------------------------------|
| Alert Priority Severity | String (Enum)  | Color-Coded Badge | No   | Values: CRITICAL (Red), WARNING (Amber), INFO (Blue) | WARNING                           |
| Alert Title             | String         | Alert Card Header | No   | Concise issue summary in regional language           | Root Zone Moisture Deficit (<35%) |
| Sensor Node Origin      | String         | Device Tag        | No   | Hardware identifier of the emitting ESP32 unit       | ESP32-NODE-01 (Plot #204)         |
| Observed Metric Value   | Float + Unit   | Numeric Metric    | No   | Physical value breaching tolerance envelope          | 31.4% VWC (Tolerance ≥ 45%)       |
| Trigger Timestamp       | DateTime (ISO) | Time Label        | No   | Time of threshold breach occurrence                  | 2026-09-16 14:22:10 IST           |
| Action Taken Status     | String (Enum)  | Status Badge      | No   | Values: UNREAD, ACKNOWLEDGED, MITIGATED              | MITIGATED (Pump Actuated)         |

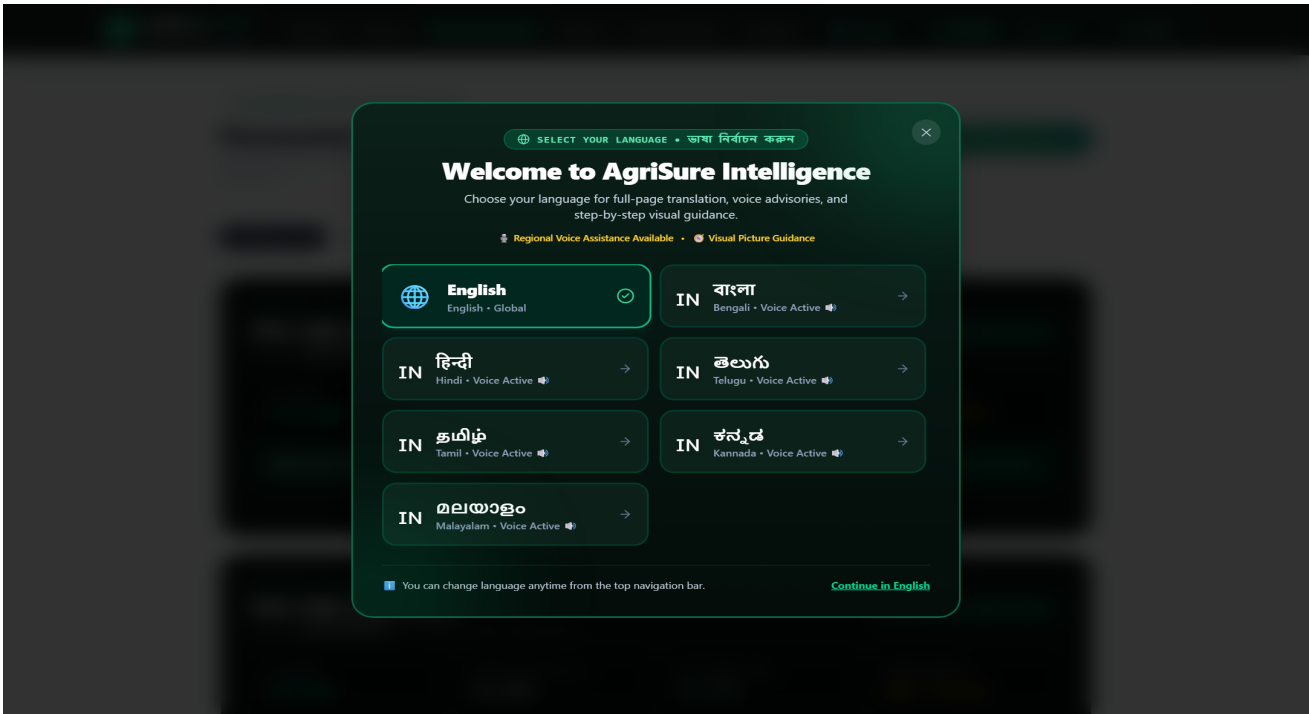
|                     |                  |               |     |                                          |                             |
|---------------------|------------------|---------------|-----|------------------------------------------|-----------------------------|
| Mitigation Shortcut | Button<br>Action | Action Button | Yes | Triggers immediate pump relay activation | Start Drip Irrigation Pulse |
|---------------------|------------------|---------------|-----|------------------------------------------|-----------------------------|

## 2.7. Parametric Crop Insurance Active Policies

Route: /farmer/insurance

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Insurance Coverage Management

Language Support: 7 Regional Languages



### Functional Purpose & UX Design:

Farmer's dedicated insurance portfolio overview. Displays all active, pending, and past policies. Highlights coverage sums insured, parametric trigger conditions (CHF breach threshold), current index safety margins, and one-click direct application for new policies.

### Data Required & Technical Pipeline:

- Farmer Policies Endpoint:** GET /api/policies?farmerId=... returns policy history, status, certificate hash, and trigger terms.
- Current CHF Monitoring:** Cross-references live plot CHF against policy trigger threshold to compute breach vigilance state.

### Field Specification & Data Schema:

| Field / Metric Name    | Data Type         | UI Control       | Req? | Validation Rules & Technical Logic                      | Sample Production Value      |
|------------------------|-------------------|------------------|------|---------------------------------------------------------|------------------------------|
| Policy Reference ID    | String (Unique)   | Card Header Code | No   | Format: POL-WB-YYYY-XXXX                                | POL-WB-2026-0881             |
| Covered Farm Plot      | String            | Plot Name Pill   | No   | Designated landholding parcel                           | Plot #204 (Baganchra, Nadia) |
| Coverage Sum Insured   | Currency (INR)    | Primary Metric   | No   | Total guaranteed parametric financial payout            | ₹1,40,000                    |
| Trigger CHF Condition  | Float (0.00-1.00) | Index Threshold  | No   | Payout triggers if 14-day mean CHF drops below          | 0.580 (Index Threshold)      |
| Current Plot Vigour    | Float (0.00-1.00) | Health Readout   | No   | Active satellite Crop Health Factor                     | 0.710 (Safe / No Breach)     |
| Policy Approval Status | String (Enum)     | Status Badge     | No   | Values: ACTIVE (Green), PENDING (Amber), REJECTED (Red) | ACTIVE & VERIFIED            |

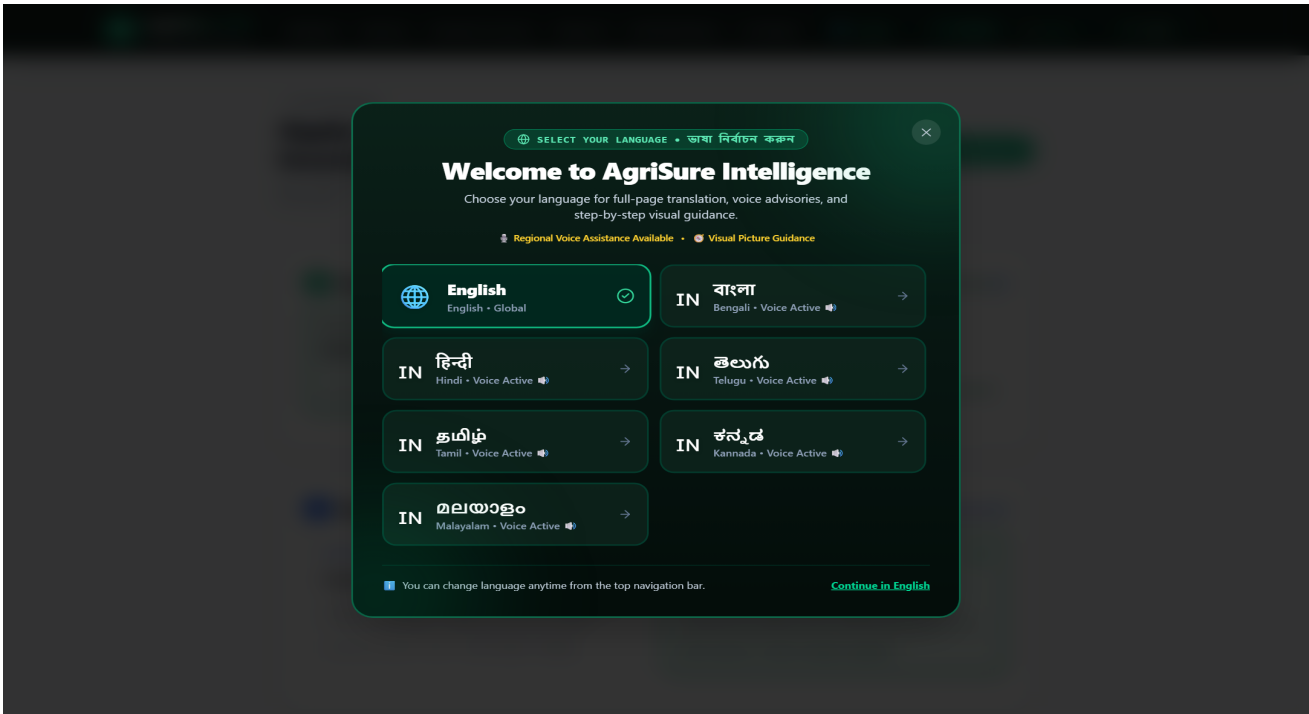
|                         |                |                     |    |                                                  |                    |
|-------------------------|----------------|---------------------|----|--------------------------------------------------|--------------------|
| Gross Premium Amount    | Currency (INR) | Financial Breakdown | No | Actuarial premium calculated for season          | ■7,000             |
| PMFBY Subsidy Credit    | Currency (INR) | Subsidy Tag         | No | 80% government financial subsidy contribution    | ■5,600 (Govt Paid) |
| Farmer Net Contribution | Currency (INR) | Net Payable         | No | Statutory 2% Kharif premium share paid by farmer | ■1,400 (Paid)      |

2.8. Guided Parametric Insurance Application Form

Route: /farmer/insurance/apply

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Financial Enrolment & Onboarding

Language Support: 7 Regional Languages



**Functional Purpose & UX Design:**

A structured 4-step guided insurance enrolment application for smallholder farmers with regional voice assistance. Features plot selection, coverage tier options (Baseline vs Comprehensive), live 80% PMFBY government subsidy calculations, automated DBT bank account verification, and instantaneous submission creating a pending underwriting record.

**Data Required & Technical Pipeline:**

- **Plot Enrolment Options:** Retrieves farmer plots eligible for Kharif 2026 coverage (Plot #204, Plot #205, Plot #212).
- **Actuarial Premium Engine:** Calculates gross premium (5%), PMFBY government subsidy (80%), and statutory farmer payable (2%).
- **DBT Bank Verification Service:** Validates linked bank account IFSC (PUNB0120111) and Aadhaar direct benefit transfer eligibility.
- **Policy Submission API:** POST /api/policies submits complete insurance application payload to institutional underwriter queue.

Field Specification & Data Schema:

| Field / Metric Name      | Data Type      | UI Control          | Req? | Validation Rules & Technical Logic                    | Sample Production Value        |
|--------------------------|----------------|---------------------|------|-------------------------------------------------------|--------------------------------|
| Selected Plot Option     | String         | Plot Radio Cards    | Yes  | Farmer selects from registered farm parcels           | Plot #204 (2.8 ha, Aman Paddy) |
| Coverage Shield Tier     | String (Enum)  | Tier Selector Cards | Yes  | Values: BASELINE (CHF 0.55), COMPREHENSIVE (CHF 0.58) | COMPREHENSIVE SHIELD           |
| Total Sum Insured        | Currency (INR) | Calculated Display  | No   | Derived: Plot Acreage × Scale of Finance (₹50k/ha)    | ₹1,40,000                      |
| Gross Premium (5%)       | Currency (INR) | Calculated Display  | No   | Standard actuarial premium rate before subsidy        | ₹7,000                         |
| PMFBY Govt Subsidy (80%) | Currency (INR) | Calculated Display  | No   | State and Central government statutory support        | -₹5,600                        |

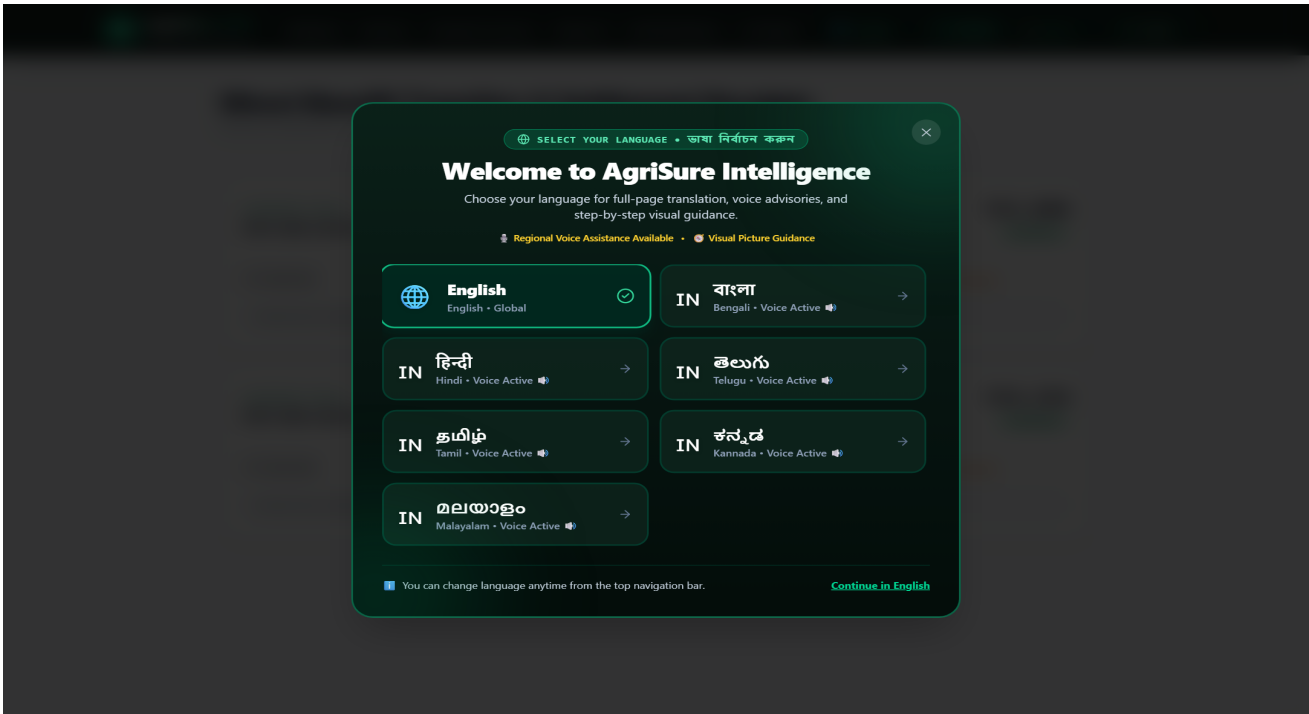
|                                      |                 |                       |     |                                                         |                                      |
|--------------------------------------|-----------------|-----------------------|-----|---------------------------------------------------------|--------------------------------------|
| <b>Farmer Net Payable (2%)</b>       | Currency (INR)  | Highlighted Cost      | No  | Actual out-of-pocket amount payable by farmer           | ■1,400                               |
| <b>Trigger CHF Threshold</b>         | Float           | Terms Specification   | No  | Index trigger breach parameter for chosen tier          | 0.580 CHF                            |
| <b>Bank Account Linkage</b>          | String (Masked) | Readonly Verification | Yes | Pre-verified bank account for automatic DBT payout      | Punjab National Bank (****8420)      |
| <b>Underwriter Application Notes</b> | String          | Text Input (Optional) | No  | Optional remarks or field history provided by applicant | Applied online via Farmer Web Portal |

2.9. Direct Benefit Bank Payouts Ledger

Route: /farmer/payments

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Financial  
Disbursements & Settlements

Language Support: 7 Regional Languages



**Functional Purpose & UX Design:**

Displays the farmer's transaction ledger of all automatic parametric claim compensations deposited directly into their bank account. Lists triggering drought/pest events, calculated loss shortfalls, RBI NEFT/IMPS transaction reference numbers, and settlement timestamps.

- Data Required & Technical Pipeline:**
- **Payouts History Endpoint:** GET /api/payouts?farmerId=... retrieves historical compensation disbursements.
  - **Banking Gateway Integration:** Simulated Reserve Bank of India National Electronic Funds Transfer (NEFT) settlement UTR records.

**Field Specification & Data Schema:**

| Field / Metric Name    | Data Type      | UI Control       | Req? | Validation Rules & Technical Logic                       | Sample Production Value               |
|------------------------|----------------|------------------|------|----------------------------------------------------------|---------------------------------------|
| Claim Sequence ID      | String         | Code Identifier  | No   | Format: CLM-YYYY-XXXX                                    | CLM-2026-0041                         |
| Peril Classification   | String         | Peril Badge      | No   | Triggering peril: DROUGHT_INDEX, EXCESS_RAIN, PEST_SHOCK | Deficit Precipitation / Drought Index |
| Index Shortfall Extent | Float (%)      | Shortfall Pill   | No   | Calculated deficit between baseline and observed CHF     | 18.5% CHF Deficit                     |
| Disbursed Compensation | Currency (INR) | Primary Amount   | No   | Direct payout credited to farmer's bank account          | ■25,900.00                            |
| Bank UTR Reference     | String         | Transaction Code | No   | Official banking settlement reference number             | NEFT-RBI-99382104                     |
| Disbursement Timestamp | DateTime       | Time Label       | No   | Timestamp of completed electronic funds transfer         | 2026-08-12 11:45:00 IST               |
| Transfer Status        | String (Enum)  | Status Badge     | No   | Values: INITIATED, TRANSFERRED_TO_BANK, SETTLED          | TRANSFERRED_TO_BANK ■                 |

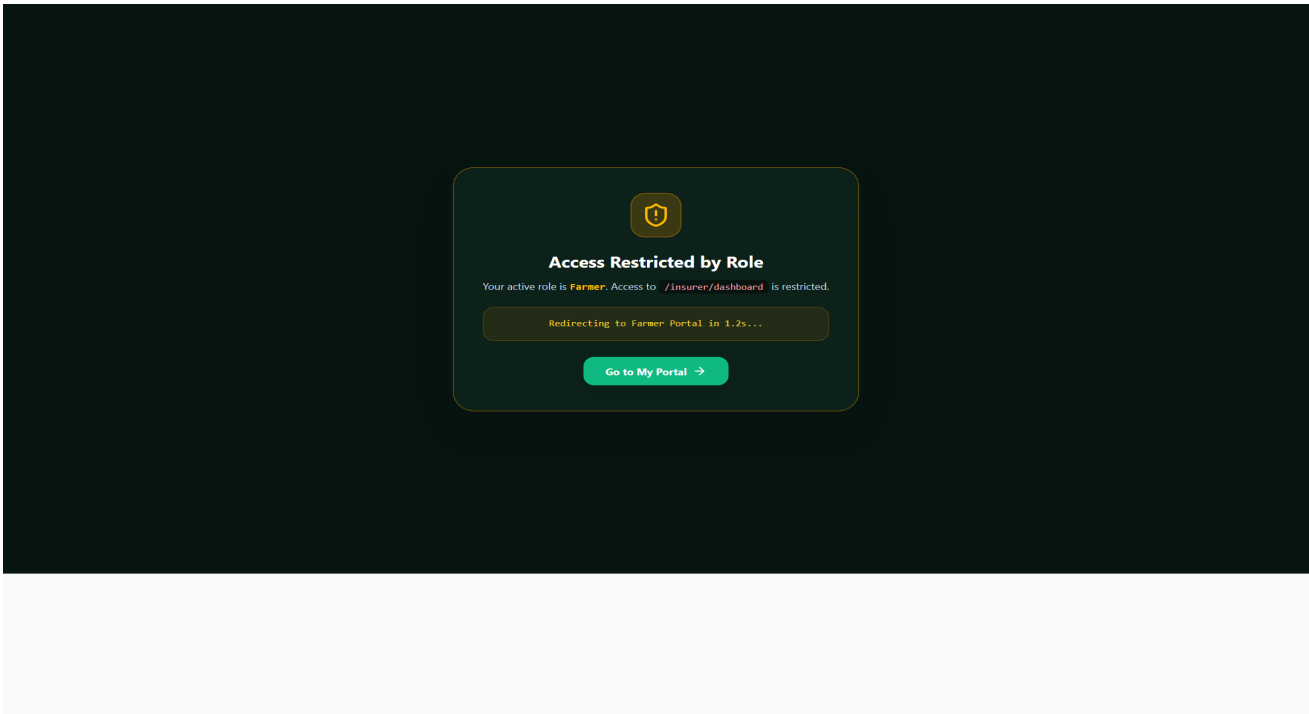


2.10. Institutional Underwriter Exposure Desk

Route: /insurer/dashboard

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Actuarial Portfolio Risk Analytics

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Macro-risk cockpit designed for insurance executives, actuaries, and reinsurers. Displays aggregate portfolio exposure across West Bengal agricultural districts, current loss ratios, active policies count, spatial CHF density distributions, flagged high-risk insurance units, and direct shortcuts to the Policy Application Review Desk.

Data Required & Technical Pipeline:

- **Portfolio Exposure Summary:** GET /api/insurance-units/summary calculates aggregate insured area (48,250 ha) and total sum insured (₹142.8 Cr).
- **Actuarial Loss Ratio:** Computes current claims payout vs gross written premium (18.4% current vs 65% ceiling).
- **Underwriting Policy Stats:** GET /api/policies/stats tracks pending application queue count and daily intake velocity.

Field Specification & Data Schema:

| Field / Metric Name      | Data Type             | UI Control    | Req? | Validation Rules & Technical Logic               | Sample Production Value |
|--------------------------|-----------------------|---------------|------|--------------------------------------------------|-------------------------|
| Total Insured Acreage    | Float (Hectares)      | Metric Card   | No   | Aggregate insured farmland under management      | 48,250 ha               |
| Total Portfolio Exposure | Currency (INR Crores) | Metric Card   | No   | Gross aggregate sum insured exposure             | ₹142.80 Crores          |
| Active Policies Enrolled | Integer               | Metric Card   | No   | Total active parametric policies issued          | 1,280 Policies          |
| Portfolio Loss Ratio     | Float (%)             | Risk Metric   | No   | Total claims paid / Total gross premium received | 18.4% (Healthy)         |
| State Mean CHF Index     | Float (0.00-1.00)     | Health Metric | No   | Mean vegetative index across all clusters        | 0.684                   |

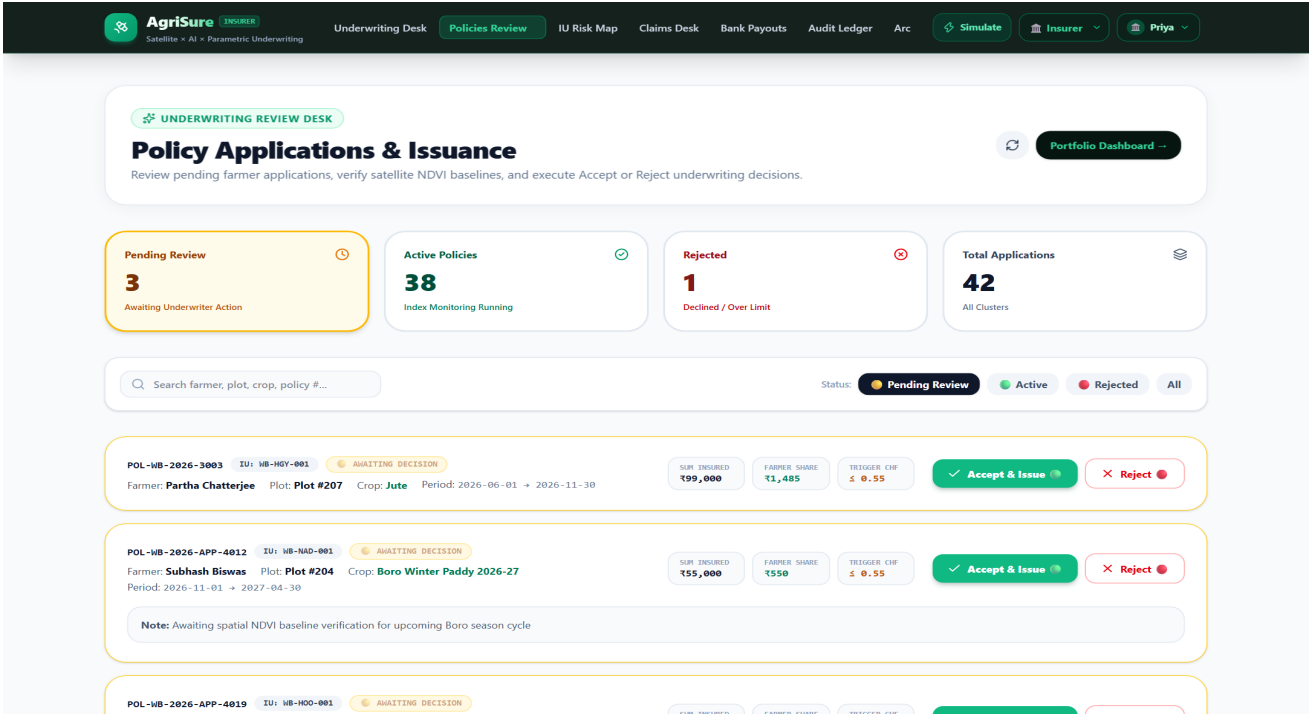
|                         |               |                     |    |                                                     |                                        |
|-------------------------|---------------|---------------------|----|-----------------------------------------------------|----------------------------------------|
| High Risk IU Units      | Integer       | Warning Alert       | No | Units with >30% plots approaching trigger threshold | 3 Units Flagged (Bankura, Murshidabad) |
| Application Queue Alert | Badge Counter | Navigation Shortcut | No | Number of pending farmer policies awaiting review   | 3 Applications Pending Review ■        |

2.11. Policy Applications & Decisioning Review Desk

Route: /insurer/policies

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Underwriting Adjudication & Governance

Language Support: English (Professional Actuarial)



**Functional Purpose & UX Design:**

Interactive underwriter review workspace where institutional officers examine incoming farmer insurance applications. Features applicant profile inspection, satellite historical baselines, and one-click 'Accept & Issue Policy' (green) or 'Reject Application' (red) with mandatory formal rejection reasons and auditable notes.

- Data Required & Technical Pipeline:**
- **Pending Applications Queue:** GET /api/policies?status=PENDING\_APPROVAL lists incoming farmer policy applications.
  - **Historical Baseline Validation:** Cross-checks plot satellite NDVI records to ensure land parcel has genuine agricultural history.
  - **Policy Decision Dispatch API:** POST /api/policies/[id]/decision with payload { decision: 'APPROVE'|'REJECT', reason, notes }.
  - **Audit Trail Inscription:** On approval or rejection, automatically commits a SHA-256 state transition block to the immutable audit ledger.

**Field Specification & Data Schema:**

| Field / Metric Name    | Data Type       | UI Control   | Req? | Validation Rules & Technical Logic       | Sample Production Value        |
|------------------------|-----------------|--------------|------|------------------------------------------|--------------------------------|
| Application ID         | String (Unique) | Table Column | No   | System identifier for policy application | pol-sub-1049                   |
| Applicant Farmer Name  | String          | Table Column | No   | Farmer legal identity                    | Rajesh Mondal / Subhash Biswas |
| Plot Identifier & Area | String + Float  | Table Column | No   | Cadastral plot identifier and acreage    | Plot #204 (2.8 ha)             |
| Crop Cultivar          | String          | Table Column | No   | Target agricultural commodity            | Aman Paddy                     |

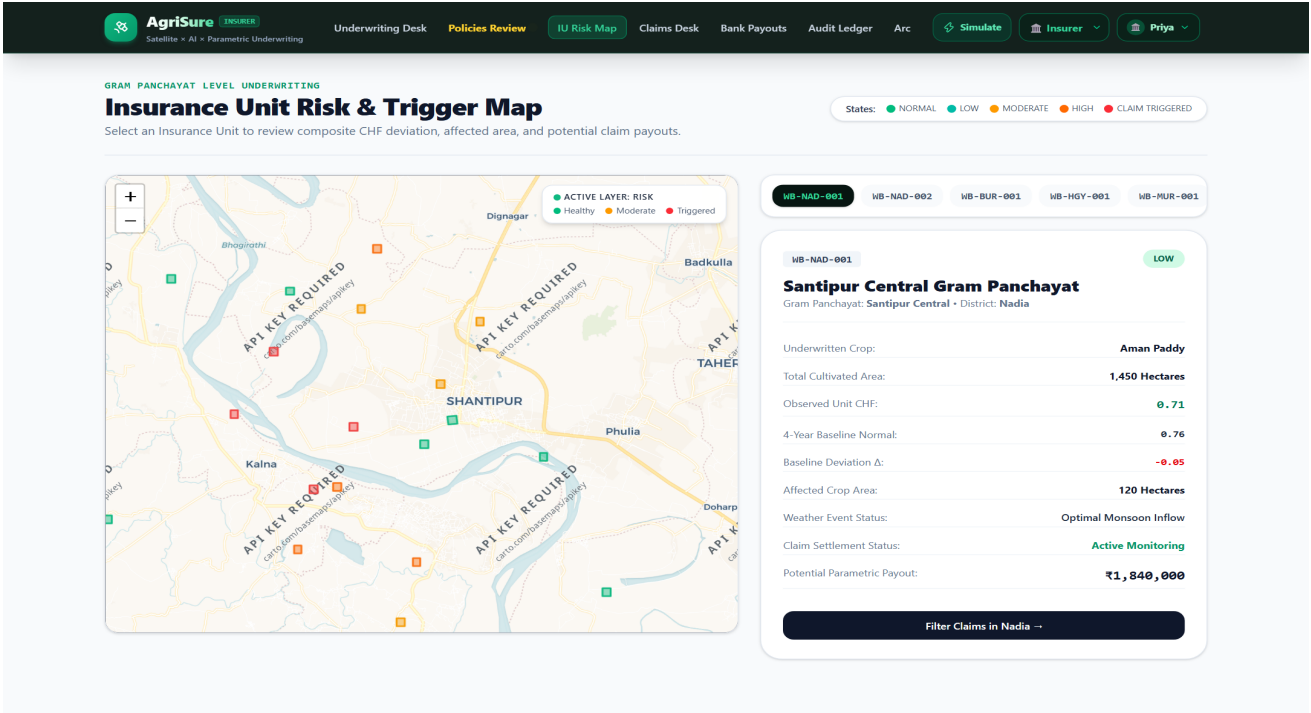
|                                 |                |                |                 |                                                          |                                              |
|---------------------------------|----------------|----------------|-----------------|----------------------------------------------------------|----------------------------------------------|
| <b>Requested Sum Insured</b>    | Currency (INR) | Table Column   | No              | Requested parametric coverage value                      | ■1,40,000                                    |
| <b>Gross Premium Calculated</b> | Currency (INR) | Table Column   | No              | Actuarial premium assessed                               | ■7,000                                       |
| <b>Govt Subsidy Credit</b>      | Currency (INR) | Table Column   | No              | 80% PMFBY financial contribution                         | ■5,600                                       |
| <b>Farmer Net Payable</b>       | Currency (INR) | Table Column   | No              | 2% farmer statutory share                                | ■1,400                                       |
| <b>Decision Action: Accept</b>  | Button (Green) | Action Button  | Yes             | Triggers policy activation and cryptographic certificate | Accept & Issue Policy ■                      |
| <b>Decision Action: Reject</b>  | Button (Red)   | Action Button  | Yes             | Opens modal for formal rejection reason selection        | Reject Application ■                         |
| <b>Formal Rejection Reason</b>  | String (Enum)  | Modal Dropdown | Yes (if reject) | Predefined IRDAI reason codes                            | EXCEEDS_SPATIAL_IU_RISK_CAP                  |
| <b>Underwriter Review Notes</b> | String         | Modal Textarea | Yes (if reject) | Auditable commentary justifying decision                 | Spatial concentration in cluster exceeds cap |

2.12. Insurance Unit Spatial Risk Choropleth Map

Route: /insurer/risk-map

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Spatial Actuarial GIS

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Geospatial choropleth console visualizing aggregated crop health, drought exposure, and parametric liability across all Insurance Units (IUs) in West Bengal. Underwriters toggle multi-spectral overlays, inspect spatial correlation matrices, and identify localized peril hotspots before threshold breach cascades.

Data Required & Technical Pipeline:

- IU GeoJSON Boundaries:** GET /api/insurance-units returns spatial polygons for WB-NAD-001, WB-BRD-004, WB-MUR-002, etc.
- Aggregated Spatial Risk Index:** Computes spatial copula model evaluating inter-plot correlation and catastrophic cluster risk.
- Raster Layer Overlay:** Integrates false-color infrared (CIR) and moisture deficit heatmaps onto the Leaflet map viewport.

Field Specification & Data Schema:

| Field / Metric Name      | Data Type         | UI Control       | Req? | Validation Rules & Technical Logic             | Sample Production Value                |
|--------------------------|-------------------|------------------|------|------------------------------------------------|----------------------------------------|
| Insurance Unit Code      | String (Unique)   | Polygon Label    | No   | Administrative unit code (e.g. WB-NAD-001)     | WB-NAD-001 (Nadia Central)             |
| District Cluster         | String            | Tooltip Header   | No   | Administrative district classification         | Nadia District                         |
| Mean Unit CHF            | Float (0.00-1.00) | Choropleth Color | No   | Green (≥0.70), Yellow (0.60-0.69), Red (<0.60) | 0.710 (Normal / Green)                 |
| Enrolled Farmland Area   | Float (ha)        | Info Sidebar     | No   | Total active insured acreage in unit           | 12,450 ha                              |
| Total Exposure Liability | Currency (INR)    | Info Sidebar     | No   | Cumulative sum insured in unit                 | ₹37.35 Crores                          |
| Drought Vulnerability    | String (Enum)     | Pill Badge       | No   | Values: LOW, MODERATE, HIGH, CRITICAL          | LOW (Adequate Monsoon)                 |
| Layer Toggles            | String (Options)  | Map Control Bar  | Yes  | NDVI, NDRE, LSWI, VH/VV Radar, Soil Moisture   | Active: Synthetic Aperture Radar (SAR) |

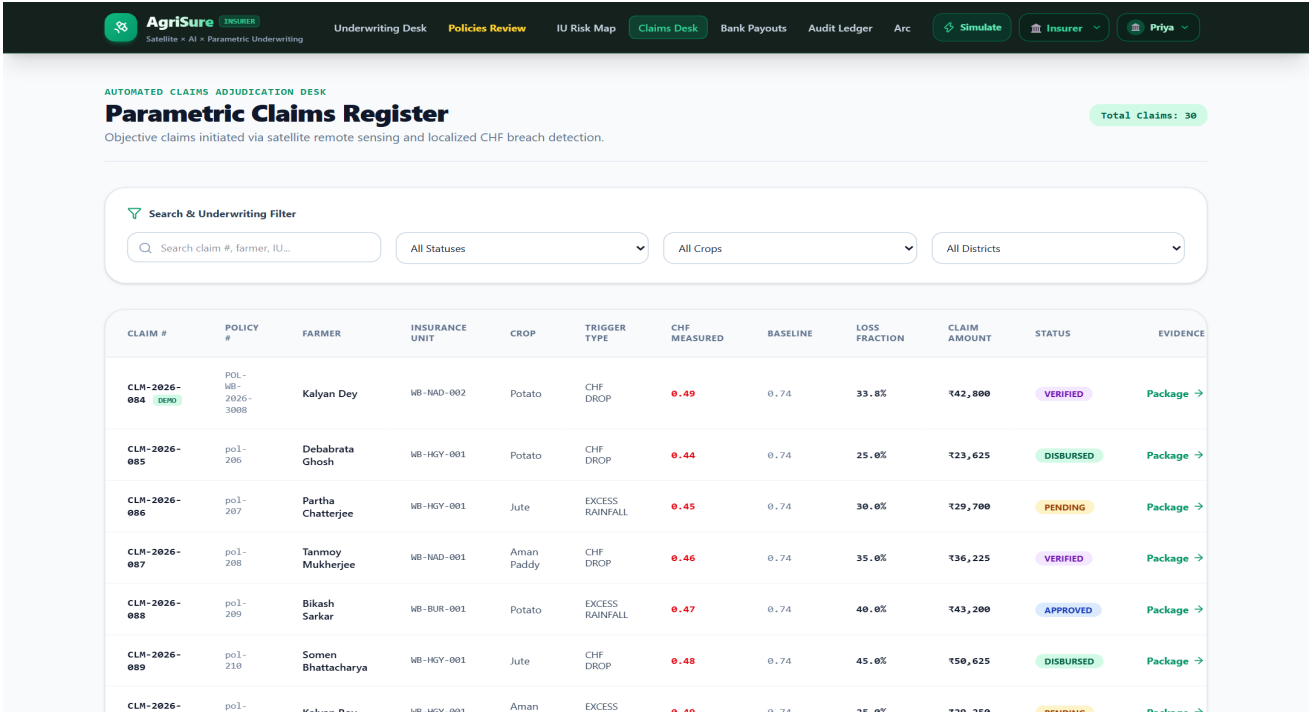


2.13. Parametric Claims Verification Console

Route: /insurer/claims

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Claims Adjudication & Evidence

Language Support: English (Professional Actuarial)



**Functional Purpose & UX Design:**

Auditing cockpit for examining automatically triggered parametric claims. Contains full 11-Section Cryptographic Evidence Dossiers compiling cadastral identity, in-situ sensor history, Sentinel-2/1 curves, Shannon entropy calculations, spatial correlation checks, and direct authorization triggers for bank disbursement.

- Data Required & Technical Pipeline:**
- **Claims Queue Endpoint:** GET /api/claims retrieves claims in VERIFIED, UNDER\_REVIEW, and SETTLED states.
  - **11-Section Dossier API:** GET /api/claims/[id] compiles complete multi-source empirical evidence package.
  - **Claim Approval API:** POST /api/claims/[id]/approve authorizes disbursement and passes claim to payments queue.

**Field Specification & Data Schema:**

| Field / Metric Name     | Data Type       | UI Control     | Req? | Validation Rules & Technical Logic            | Sample Production Value     |
|-------------------------|-----------------|----------------|------|-----------------------------------------------|-----------------------------|
| Claim Identification ID | String (Unique) | Table Column   | No   | Format: CLM-YYYY-XXXX                         | CLM-2026-0041               |
| Policy Reference        | String          | Linked Code    | No   | Underlying parametric policy contract         | POL-WB-2026-0881            |
| Beneficiary Farmer      | String          | Table Column   | No   | Claimant farmer name                          | Rajesh Mondal               |
| Plot Identifier         | String          | Table Column   | No   | Affected farmland plot                        | Plot #204 (Baganchra)       |
| Triggering Peril        | String (Enum)   | Peril Badge    | No   | Values: DROUGHT_INDEX, FLOOD_SAR, HEAT_STRESS | DROUGHT_INDEX (CHF Deficit) |
| Trigger CHF Breached    | Float           | Numeric Column | No   | Contractually specified breach threshold      | 0.580 CHF                   |
| Observed Mean CHF       | Float           | Numeric Column | No   | Actual 14-day observed composite index        | 0.472 CHF (Deficit 18.5%)   |

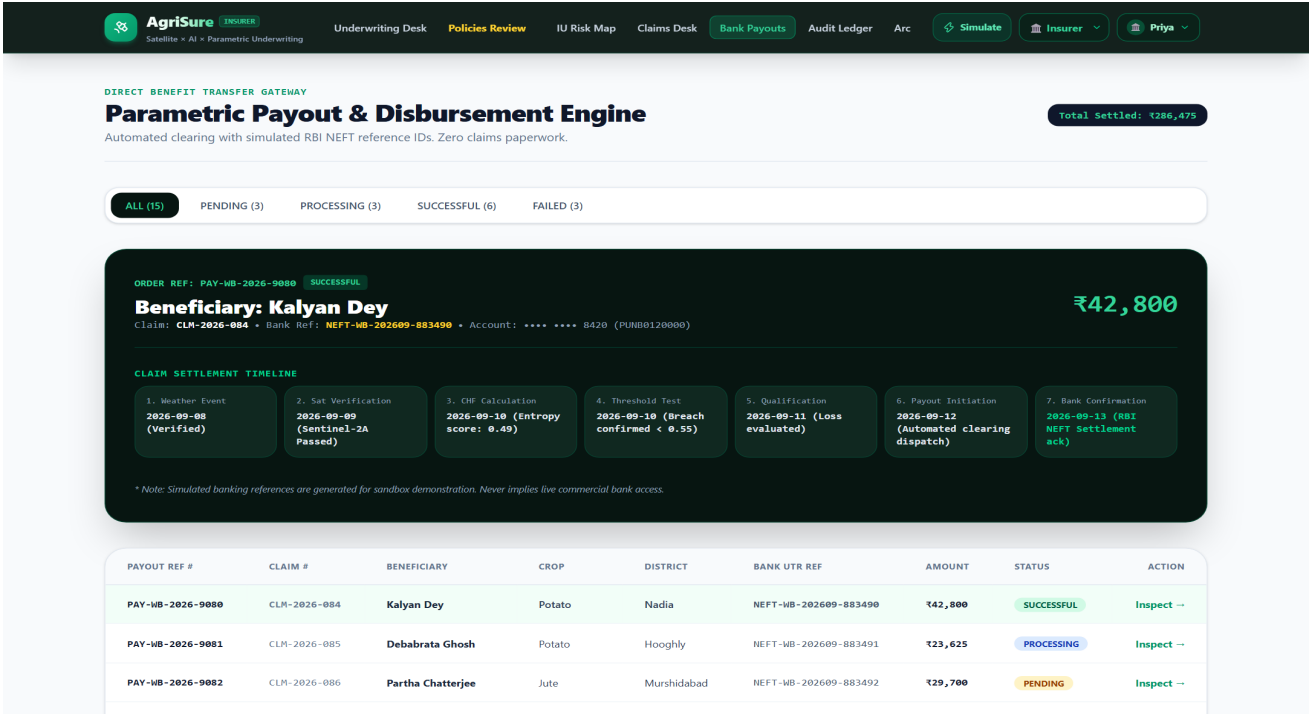
|                          |                  |                  |     |                                                      |                           |
|--------------------------|------------------|------------------|-----|------------------------------------------------------|---------------------------|
| Assessed Payout Amount   | Currency (INR)   | Financial Column | No  | Compensation determined by index shortfall formula   | ■25,900.00                |
| Evidence Package Status  | String           | Status Badge     | No  | Verification of all 11 scientific evidence sections  | 11/11 SECTIONS VERIFIED ■ |
| Action: Authorize Payout | Button (Emerald) | Action Button    | Yes | Directs claim to automated RBI disbursement pipeline | Authorize Payout →        |

2.14. 7-Stage Automated RBI NEFT/IMPS Settlement

Route: /insurer/payouts

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Banking & Funds Disbursement

Language Support: English (Professional Actuarial)



**Functional Purpose & UX Design:**

Visual workflow tracker monitoring the 7 automated settlement stages: 1. Trigger Detection → 2. Spatial Cross-Validation → 3. Quantum Determination → 4. Reinsurance Allocation → 5. Underwriter Authorization → 6. RBI Gateway Transmission → 7. DBT Account Credit. Features simulated RBI NEFT/IMPS settlement execution.

- Data Required & Technical Pipeline:**
- **Settlements Batch Endpoint:** GET /api/payouts lists queued and executed bank direct transfer transactions.
  - **Instant Settlement Trigger:** POST /api/payouts/[id]/execute dispatches funds transfer to simulated RBI gateway.

**Field Specification & Data Schema:**

| Field / Metric Name     | Data Type        | UI Control              | Req? | Validation Rules & Technical Logic               | Sample Production Value           |
|-------------------------|------------------|-------------------------|------|--------------------------------------------------|-----------------------------------|
| Payout Batch Reference  | String (Unique)  | Batch Header            | No   | Format: PAY-BATCH-YYYY-MM-XXXX                   | PAY-BATCH-2026-09-001             |
| Disbursement Amount     | Currency (INR)   | Primary Card Metric     | No   | Total funds released in this transaction         | ■25,900.00                        |
| Beneficiary Bank Name   | String           | Bank Badge              | No   | Farmer's registered direct benefit transfer bank | Punjab National Bank              |
| Beneficiary IFSC Code   | String           | Readonly Code           | No   | Branch IFSC code (11 characters)                 | PUNB0120111                       |
| Settlement Stage Active | Integer (1 to 7) | Step Progress Indicator | No   | Active pipeline stage of the 7 settlement steps  | Stage 6: RBI Gateway Transmission |

|                     |               |                    |    |                                                |                           |
|---------------------|---------------|--------------------|----|------------------------------------------------|---------------------------|
| RBI UTR Number      | String        | Receipt Identifier | No | Unique Transaction Reference from central bank | NEFT-RBI-99382104         |
| Execution Timestamp | DateTime      | Time Record        | No | Settlement execution time                      | 2026-09-16 11:45:00 IST   |
| Settlement State    | String (Enum) | Status Badge       | No | Values: QUEUED, IN_TRANSIT, COMPLETED          | COMPLETED & CREDITED<br>■ |

2.15. Immutable Cryptographic SHA-256 Audit Trail

Route: /insurer/audit

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Integrity & Compliance Ledger

Language Support: English (Professional Actuarial)

AgriSureINSURER

Satellite • AI • Parametric Underwriting

Underwriting DeskPolicies ReviewIU Risk MapClaims DeskBank PayoutsAudit LedgerArc

SimulateInsurerPriya

CRYPTOGRAPHIC EVIDENCE LEDGER

Immutable Underwriting Audit Trail

Every satellite scene, model execution, threshold evaluation, and banking dispatch is anchored with SHA-256 block hashes.

CHAIN INTEGRITY: VERIFIED

Search entity ID, action, actor...

All Entity Types

PARAMETRIC\_EVALUATION\_PASSED • CLM-2026-084

2026-09-15 10:14:22 UTC • Actor: CHF-Automated-Engine-v2.4

Scene ID  
S2A\_MSL2A\_20260912

Satellite Source  
Sentinel-2A MSI / Sentinel-1B SAR

Acquisition Time  
2026-09-12 04:46 UTC

Weather Source / Ver  
IMD AWS Station 4208 (v2.1)

Model Version  
CropYield-CHF-v2.4

CHF Engine Version  
Entropy-Norm-v2.4

Baseline Version  
Nadia-Aman-4Yr-Norm-2026

Decision Rule  
CHF <= 0.55 THRESHOLD TRIGGER

IoT Readings  
SoilMoisture=88.5%, R=420ohm

Ground Evidence  
FA-MB-441 GPS (23.161, 88.542)

Bank UTR Ref  
NEFT-MB-202609-883492

Ledger Status  
VALID\_CHAIN

SHA-256 Block Hash: 9f83a42cbe8192a7493d8fb654128f1190bc1294817e02c8942b012354f9bc41

Chain Validated • Tamper Evident

PAYOUT\_DISPATCH\_AUTHORIZED • PAY-MB-2026-9084

2026-09-14 16:45:10 UTC • Actor: Underwriter:Dr. S. Sen (Chief Actuary)

Scene ID  
S2A\_MSL2A\_20260912

Satellite Source  
Sentinel-2A

Acquisition Time  
2026-09-12 04:46 UTC

Weather Source / Ver  
IMD AWS Station 4208 (v2.1)

Model Version  
MobileNetV3-Agronomic-v3.2

CHF Engine Version  
Entropy-Norm-v2.4

Baseline Version  
Nadia-Aman-4Yr-Norm-2026

Decision Rule  
Approved per Loss Ratio 0.338

IoT Readings  
SoilMoisture=88.5%, R=420ohm

Ground Evidence  
FA-MB-441 GPS Verified

Bank UTR Ref  
NEFT-MB-202609-883492

Ledger Status  
VALID\_CHAIN

SHA-256 Block Hash: a127f884210ecba1950482da9b7468194481028751992043681423859102431f

Chain Validated • Tamper Evident

Functional Purpose & UX Design:

Enterprise cryptographic audit ledger recording every state change, satellite pass calculation, model inference, underwriter decision, and banking settlement in a chained SHA-256 Merkle structure. Guarantees regulatory transparency for IRDAI auditors and reinsurers.

Data Required & Technical Pipeline:

- Audit Blocks Endpoint:** GET /api/audit lists chained transaction blocks with cryptographic hash verification.
- Merkle Root Computation:** Calculates cryptographic root hash certifying integrity against previous block hash.

Field Specification & Data Schema:

| Field / Metric Name | Data Type        | UI Control     | Req? | Validation Rules & Technical Logic                           | Sample Production Value           |
|---------------------|------------------|----------------|------|--------------------------------------------------------------|-----------------------------------|
| Audit Block Number  | Integer          | Block Header   | No   | Sequential immutable block index                             | Block #0842                       |
| Timestamp (UTC)     | DateTime         | Time Label     | No   | ISO-8601 millisecond-precision timestamp                     | 2026-09-16T14:22:10.842Z          |
| Entity Type         | String (Enum)    | Entity Pill    | No   | Values: POLICY, CLAIM, SATELLITE_PASS, TELEMETRY, PAYOUT     | POLICY                            |
| Event Action        | String           | Action Label   | No   | Operation executed (e.g. POLICY_ISSUED, PUMP_TOGGLED)        | POLICY_UNDERWRITING_APPROVED      |
| Actor Principal     | String           | Actor Tag      | No   | Entity executing transaction (Underwriter ID, System Oracle) | Priya Sengupta (IRDAI-AGR-8842)   |
| Previous Block Hash | SHA-256 (64 hex) | Code Monospace | No   | Cryptographic hash of immediately preceding block            | 7f83b1657ff1fc53b92dc18148a1d6... |
| Current Merkle Hash | SHA-256 (64 hex) | Code Monospace | No   | SHA-256 hash of current block payload                        | a591a6d40bf420404a011733cfb7b1... |

|                     |        |                |    |                                    |                                 |
|---------------------|--------|----------------|----|------------------------------------|---------------------------------|
| Verification Status | String | Security Badge | No | Chain integrity verification check | CRYPTOGRAPHICALLY<br>VERIFIED ■ |
|---------------------|--------|----------------|----|------------------------------------|---------------------------------|

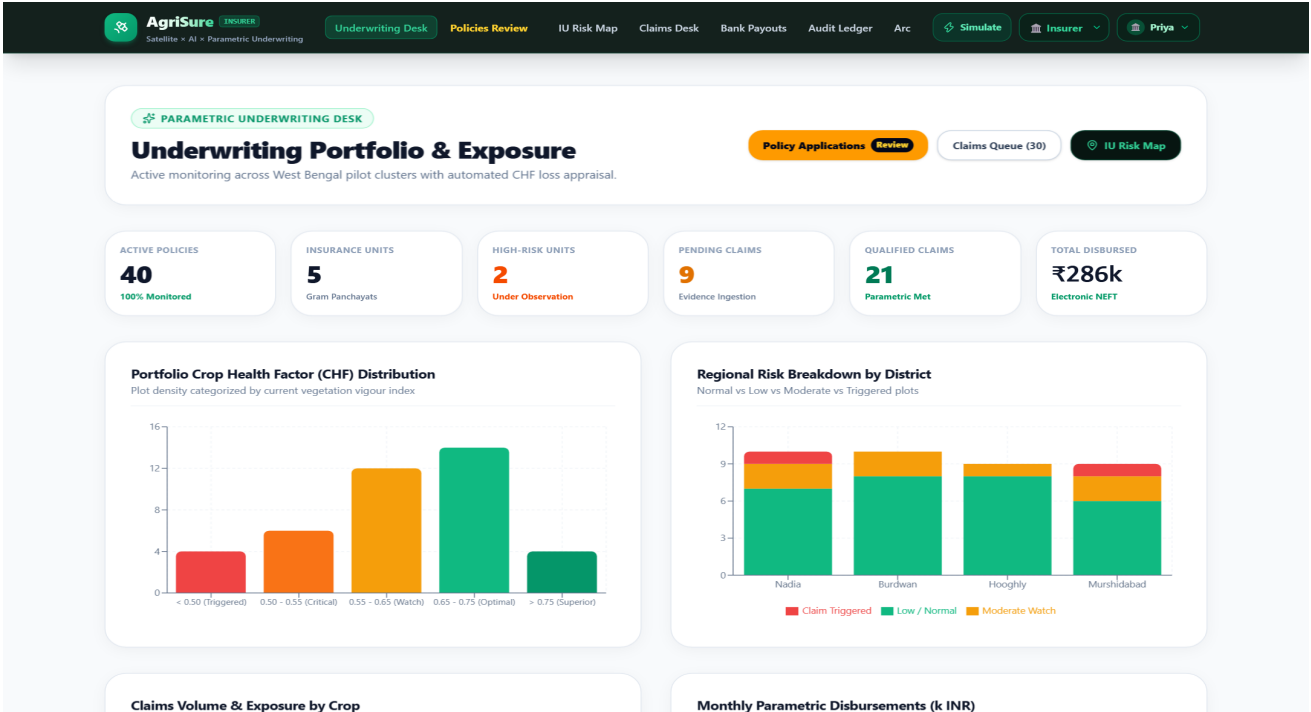
2.16. Hardware IoT Ingestion & Pump Relay Control

Route: /operations/iot

Authorized RBAC Persona: ADMIN & FIELD AGENT | Tier: Hardware Infrastructure

Language Support: English

Operations



Functional Purpose & UX Design:

Operations control room monitoring the network of 20 field IoT stations across West Bengal. Displays hardware heartbeat, battery voltage (LiFePO4 4.1V), cellular/LoRa signal RSSI, dual-probe capacitance calibration, and bi-directional pump relay toggles.

Data Required & Technical Pipeline:

- **IoT Fleet Endpoint**: GET /api/iot/devices retrieves status and readings across all 20 hardware nodes.
- **Remote Actuation Dispatch**: POST /api/iot/devices/[id]/toggle pulses hardware GPIO relay on physical device.
- **Heartbeat Tick Monitor**: POST /api/iot/tick simulates streaming packet arrivals from field edge gateways.

Field Specification & Data Schema:

| Field / Metric Name         | Data Type       | UI Control      | Req? | Validation Rules & Technical Logic            | Sample Production Value             |
|-----------------------------|-----------------|-----------------|------|-----------------------------------------------|-------------------------------------|
| Device Hardware ID          | String (Unique) | Card Header     | No   | MAC / IMEI hardware device tag                | ESP32-NODE-01 (Plot #204)           |
| Installed Plot Location     | String          | Location Pill   | No   | Farmland installation site                    | Baganchra, Santipur Central (Nadia) |
| Battery Storage Voltage     | Float (Volts)   | Power Indicator | No   | LiFePO4 solar battery voltage: 3.2V to 4.2V   | 4.12 V (Solar Inflow Active)        |
| Cellular / LoRa RSSI        | Integer (dBm)   | Signal Bar      | No   | Signal strength: optimal is > -85 dBm         | -78 dBm (Strong Link)               |
| Root Soil Moisture (15cm)   | Float (%)       | Sensor Metric   | No   | Volumetric water content in primary root zone | 58.0% VWC                           |
| Sub-surface Moisture (30cm) | Float (%)       | Sensor Metric   | No   | Deep capacitance moisture reading             | 64.5% VWC                           |
| Soil Temperature            | Float (°C)      | Sensor Metric   | No   | Direct in-ground probe temperature            | 28.4 °C                             |

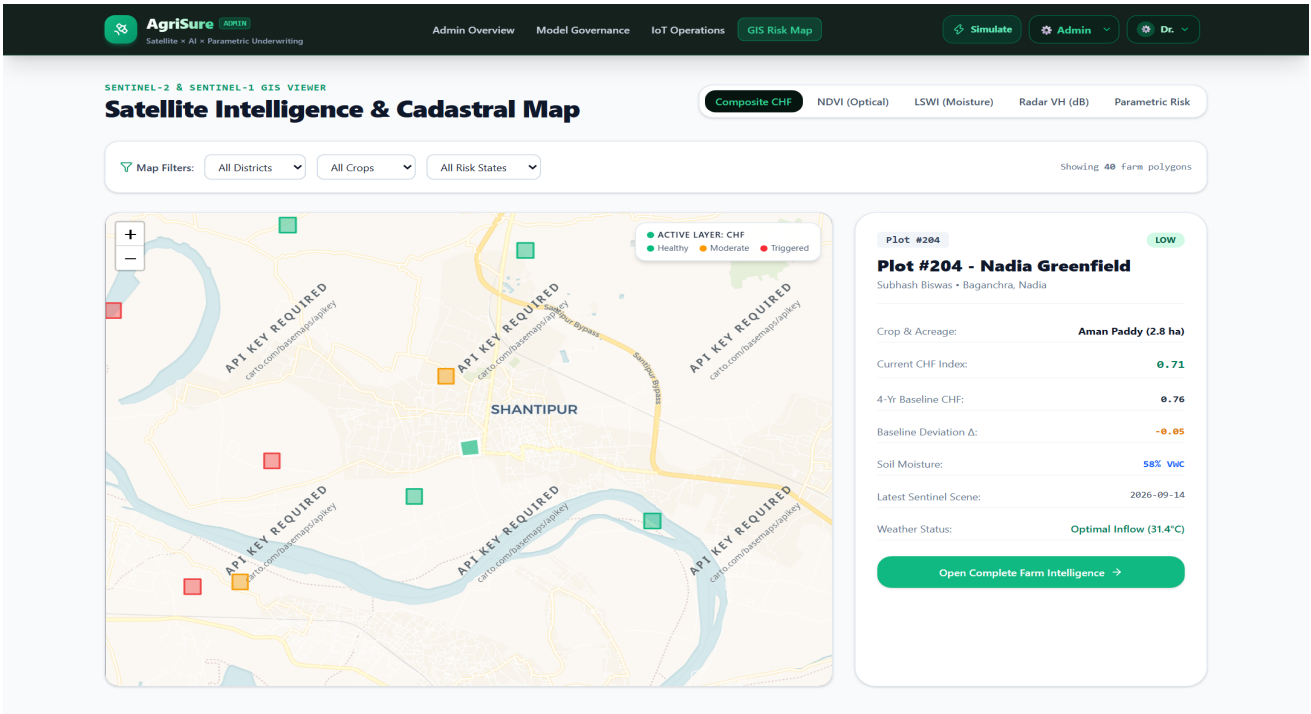
|                  |               |                |     |                                                 |                        |
|------------------|---------------|----------------|-----|-------------------------------------------------|------------------------|
| Pump Relay State | Boolean       | Toggle Control | Yes | Physical relay contact: CLOSED (1) or OPEN (0)  | RELAY ACTIVE (Pump On) |
| Flowmeter Rate   | Float (L/min) | Flow Metric    | No  | Discharge volume through inline micro-drip line | 42.5 L/min             |

2.17. Sentinel Multispectral GIS Analytics Console

Route: /intelligence/map

Authorized RBAC Persona: ADMIN & FIELD AGENT | Tier: Geospatial Earth Observation

Language Support: English



Functional Purpose & UX Design:

High-resolution spatial GIS explorer overlaying European Space Agency Sentinel-2 optical multispectral imagery and Sentinel-1 C-band SAR radar data. Users toggle false color infrared composites, inspect pixel-level vegetative indices, and analyze cloud penetration.

Data Required & Technical Pipeline:

- **Satellite Raster Tile Service:** Streams Sentinel-2 L2A bottom-of-atmosphere reflectance tiles and Sentinel-1 SAR GRD tiles.
- **Cloud Masking Engine:** Applies Scene Classification Layer (SCL) to filter atmospheric haze and cloud shadows.
- **Spatial Query Point Intersect:** Allows clicking any pixel coordinate to retrieve full spectral signature (B2-B12).

Field Specification & Data Schema:

| Field / Metric Name       | Data Type         | UI Control        | Req? | Validation Rules & Technical Logic                         | Sample Production Value           |
|---------------------------|-------------------|-------------------|------|------------------------------------------------------------|-----------------------------------|
| Spectral Band Preset      | String (Enum)     | Layer Dropdown    | Yes  | RGB True Color, CIR Infrared, SWIR Agriculture             | Agriculture Composite (B11-B8-B2) |
| Active Remote Index       | String (Enum)     | Index Selector    | Yes  | NDVI, NDRE, EVI, LSWI, Radar VH/VV Ratio                   | NDVI (Normalized Difference Veg)  |
| Cloud Mask Threshold      | Integer (%)       | Slider Control    | Yes  | Maximum acceptable scene cloud percentage                  | 15% Maximum Cloud Cover           |
| Pixel Cursor Coordinate   | Float (Lat/Lon)   | Status Coordinate | No   | Geographic location of viewport cursor                     | LAT 23.472°N • LON 88.556°E       |
| Sensor Satellite Platform | String            | Metadata Pill     | No   | Active spacecraft: Sentinel-2A / Sentinel-2B / Sentinel-1A | Sentinel-2B MSI (10m Resolution)  |
| Pixel CHF Health Score    | Float (0.00-1.00) | Inspect Tooltip   | No   | Calculated vegetative health at sampled point              | 0.710 (High Vigour)               |

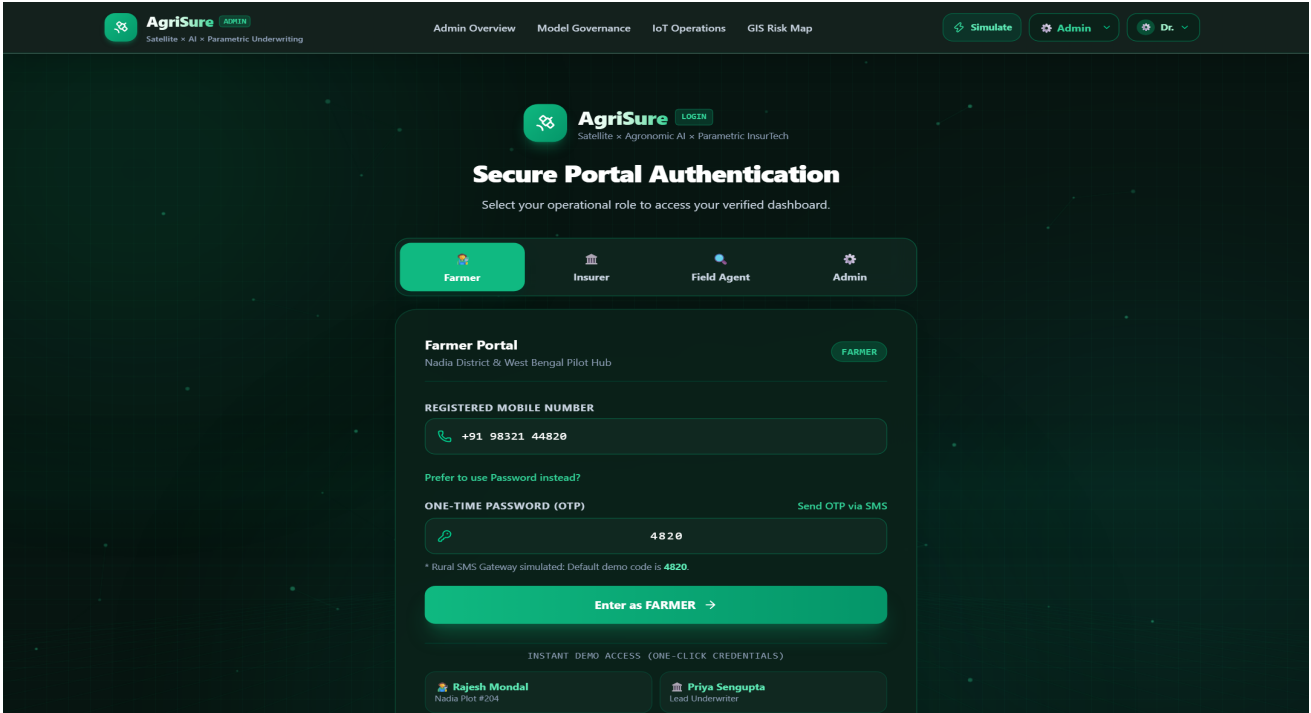


2.18. Multi-Role Authentication & OTP Portal

Route: /login

Authorized RBAC Persona: Public Universal | Tier: Access Control & Identity

Language Support: English / Role Adaptive



Functional Purpose & UX Design:

Secure entry portal supporting four distinct personas: Farmer, Insurer, System Administrator, and Field Agent. Provides 1-click evaluation presets for instant demonstration, SMS OTP authentication for rural farmers, and credential login for underwriters.

Data Required & Technical Pipeline:

- **Authentication API:** POST /api/auth/login validates credentials, issues auth token, and sets agrisure\_user\_role cookie.
- **SMS OTP Service:** Simulated OTP dispatch to farmer's mobile number (+91 98321 44820).

Field Specification & Data Schema:

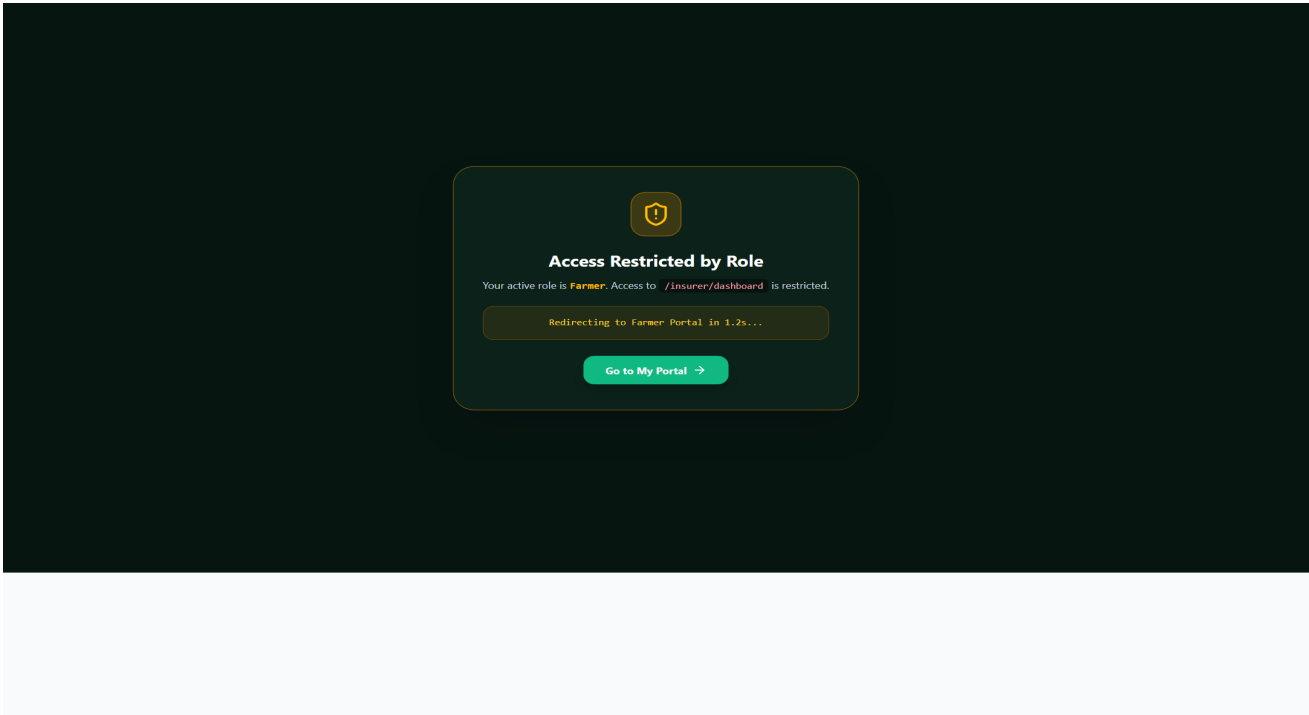
| Field / Metric Name | Data Type       | UI Control           | Req? | Validation Rules & Technical Logic                 | Sample Production Value                  |
|---------------------|-----------------|----------------------|------|----------------------------------------------------|------------------------------------------|
| Target Role Persona | String (Enum)   | Role Selector Tabs   | Yes  | Values: FARMER, INSURER, ADMIN, FIELD_AGENT        | Farmer                                   |
| User Identifier     | String          | Text Input           | Yes  | Mobile Number (Farmers) or Work Email (Insurers)   | +91 98321 44820 / p.sengupta@agrisure.in |
| Password / SMS OTP  | String (Secret) | Password / OTP Input | Yes  | Password or 6-digit one-time SMS verification code | / 842011                                 |
| Instant Demo Preset | Button Action   | Quick-Fill Buttons   | No   | One-click autofill of verified persona credentials | Autofill Rajesh Mondal (Farmer)          |
| Keep Logged In      | Boolean         | Checkbox             | No   | Extends session cookie lifespan to 30 days         | TRUE (Checked)                           |

2.19. 6-Stage Technical Architecture & Agronomic Math

Route: /how-it-works

Authorized RBAC Persona: INSURER & ADMIN | Tier: Technical Documentation

Language Support: English



**Functional Purpose & UX Design:**

Comprehensive technical specification detailing the mathematical foundations: the Shannon Entropy Crop Health Factor formula, GA-PSO bio-inspired heuristic equations, synthetic aperture radar backscatter processing, copula spatial risk aggregation, and the complete 6-stage operational protocol.

**Data Required & Technical Pipeline:**

- **Mathematical Specifications:** LaTeX-rendered equations for Shannon entropy sub-index weights and GA-PSO fitness functions.
- **Agronomic Protocol Validation:** Citations of university research models benchmarked against West Bengal agro-ecological zones.

**Field Specification & Data Schema:**

| Field / Metric Name     | Data Type | UI Control     | Req? | Validation Rules & Technical Logic                                     | Sample Production Value                  |
|-------------------------|-----------|----------------|------|------------------------------------------------------------------------|------------------------------------------|
| Shannon Entropy Formula | Equation  | Math Block     | No   | $CHF = \sum w_i \times I_i$ where $w_i$ is derived via entropy weights | CHF Formula Matrix                       |
| GA-PSO Fitness Function | Equation  | Math Block     | No   | Minimizes cost & nutrient leaching; maximizes vegetative vigour        | Fitness Objective Equation               |
| SAR Polarimetric Ratio  | Equation  | Math Block     | No   | VH/VV cross-polarization backscatter ratio for soil moisture           | $\gamma = \sigma^0_{VH} / \sigma^0_{VV}$ |
| Parametric Trigger Spec | Rule Spec | Protocol Block | No   | Threshold breach specification and claim qualification logic           | $CHF \leq 0.580$ for 14 consecutive days |

### 3. Database Schema & Data Dictionary (Prisma ORM)

The system persists all agro-financial state across 22 PostgreSQL models defined in prisma/schema.prisma with PostGIS geometry support. Below is the operational data dictionary of the primary core entities:

| Entity Model     | Primary Key | Key Backing Attributes                                                                                   | Operational Role in Platform                                                             |
|------------------|-------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| User             | id (UUID)   | name, role (Enum), phone, email, district, village, bankAccount                                          | Authenticates Farmers, Insurers, Admins, and Agents with RBAC isolation.                 |
| Farm             | id (UUID)   | farmerId, plotNumber, areaHa, crop, boundaryGeoJson, iuCode                                              | Cadastral plot record with spatial boundaries for satellite intersection.                |
| IoTDevice        | id (UUID)   | farmId, deviceCode, batteryV, rssiDbm, pumpRelayActive, lastPing                                         | Hardware telemetry ingestion node and remote micro-irrigation pump relay.                |
| SensorReading    | id (UUID)   | deviceId, soilMoisture15, soilMoisture30, soilTemp, ambientRh, recordedAt                                | High-frequency 3s timeseries tracking volumetric root zone water content.                |
| CropHealthRecord | id (UUID)   | farmId, chfIndex, ndvi, ndre, lswi, sarBackscatter, sourceDate                                           | Computed Shannon entropy composite vigour derived from Sentinel-2/1 passes.              |
| Policy           | id (UUID)   | policyNumber, farmId, farmerId, sumInsured, grossPremium, subsidyAmount, farmerShare, triggerChf, status | Smart parametric contract binding farmer, underwriter, and index trigger threshold.      |
| Claim            | id (UUID)   | claimNumber, policyId, perilType, triggerDate, deficitChf, payoutAmount, status, evidenceDossierJson     | Automated claim qualification record backed by 11-section empirical evidence package.    |
| Payout           | id (UUID)   | claimId, farmerId, amount, bankName, ifsc, utrNumber, stage (1-7), settledAt                             | Direct Benefit Transfer ledger tracking electronic funds settlement to farmer's account. |
| AuditBlock       | id (UUID)   | blockNumber, entityType, actorId, prevHash, merkleHash, payloadJson, createdAt                           | Chained SHA-256 Merkle ledger certifying immutable platform transparency.                |

## 4. REST API Endpoint Inventory & Edge Security

All API routes are served from Next.js 16 Route Handlers (`src/app/api/*`) under strict HTTP method conventions and edge proxy guard verification:

| Method | Endpoint Path                             | Required Role    | Data Payload & Response Summary                                                         |
|--------|-------------------------------------------|------------------|-----------------------------------------------------------------------------------------|
| POST   | <code>/api/auth/login</code>              | Universal        | Receives credentials/OTP; sets <code>agrisure_user_role</code> cookie; returns profile. |
| POST   | <code>/api/auth/register</code>           | Universal        | Creates farmer or institutional user account; stores initial profile record.            |
| GET    | <code>/api/farms</code>                   | FARMER, AGENT    | Returns list of registered farms with GeoJSON boundaries and vegetative scores.         |
| GET    | <code>/api/farms/[id]/sensors</code>      | FARMER, AGENT    | Returns real-time soil moisture (15cm & 30cm), temp, and ambient humidity.              |
| POST   | <code>/api/disease/predict</code>         | FARMER           | Multipart leaf image → 38-class CV inference, confidence, and spray remedy.             |
| POST   | <code>/api/optimization/run</code>        | FARMER           | Soil N-P-K baseline → GA-PSO fertilizer (Urea, DAP, MOP) and water schedule.            |
| POST   | <code>/api/iot/devices/[id]/toggle</code> | FARMER, ADMIN    | Toggles physical pump relay between ACTIVE (1) and STANDBY (0).                         |
| GET    | <code>/api/policies</code>                | Universal        | Lists policies filtered by status (ACTIVE, PENDING_APPROVAL, REJECTED).                 |
| POST   | <code>/api/policies</code>                | FARMER           | Submits new insurance application with subsidy calculation and bank verification.       |
| POST   | <code>/api/policies/[id]/decision</code>  | INSURER          | Accepts or rejects application with formal reason and audit ledger inscription.         |
| GET    | <code>/api/claims</code>                  | INSURER          | Retrieves all claims with 11-section empirical cryptographic evidence dossiers.         |
| POST   | <code>/api/claims/[id]/approve</code>     | INSURER          | Approves qualified parametric claim and queues it for automated bank disbursement.      |
| GET    | <code>/api/payouts</code>                 | Universal        | Lists 7-stage automated bank compensation transfers with simulated RBI UTRs.            |
| POST   | <code>/api/payouts/[id]/execute</code>    | INSURER          | Executes instantaneous electronic direct benefit deposit to farmer's bank account.      |
| GET    | <code>/api/audit</code>                   | INSURER, ADMIN   | Streams immutable chained SHA-256 Merkle audit blocks for regulatory compliance.        |
| POST   | <code>/api/simulate/event</code>          | Universal (Demo) | Injects synthetic weather shock: Severe Drought, Flash Flood, or Foliar Blight.         |

### Deployment Topology & Production Runtime:

- **Vercel Production Edge:** Hosts the Next.js 16 App Router frontend with Turbopack acceleration, regional edge middleware route gating, and optimized static asset delivery.
- **Render Cloud Platform:** Hosts the containerized database layer (PostgreSQL 16 with PostGIS spatial extension) and telemetry background daemon workers.
- **Edge Route Protection (`src/middleware.ts`):** Inspects incoming `agrisure_user_role` cookies. Blocks and redirects unauthorized cross-role route attempts (e.g. Farmers cannot visit `/insurer/*` or `/admin/*`; Insurers cannot visit `/farmer/*`).
- **Client-Side Watchdog (`RouteGuard.tsx`):** Intercepts client navigation and displays an animated modal redirecting the user to their authorized dashboard within 1.2 seconds.